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HEALTH AND MEDICINE

AI-driven discovery of dual antiaging and anti-AD therapeutics via PROTAC target deconvolution of a super-enhancer–regulated axis

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The lack of safe, durable therapeutics that act against both biological aging and Alzheimer's disease is an unmet clinical need. To bridge this gap, we devised an artificial intelligence (AI)–enabled approach that pairs rapid compound triage with mechanistic target deconvolution. Our AI-driven screening highlighted melatonin (MLT) as a promising candidate. Serum profiling of 161 human individuals confirmed an age-related fall in circulating MLT level, while subsequent in vivo and in vitro experiments showed that MLT rescues cognition, suppresses neuroinflammation, and alleviates senescence phenotypes. Proteolysis targeting chimera (PROTAC)–guided chemoproteomic deconvolution next pinpointed the histone acetyltransferase p300 as MLT's target. Integrated Cleavage Under Targets and Tagmentation, single-cell RNA sequencing, and spatial transcriptomics revealed that MLT-bound p300 cooperates with specificity protein 1 (SP1) at a brain and muscle ARNT-like protein 1 super-enhancer, elevating histone H3 lysine-27 acetylation and reengaging a circadian-epigenetic program that links redox resilience to neuroprotection. By combining AI-driven discovery with PROTAC-based target mapping and super-enhancer–centric mechanistic resolution, our study identifies MLT as a dual-action candidate and sets out a reproducible "AI-to-clinic" paradigm for multitarget drug innovation in aging-related neurodegeneration.

INTRODUCTION

Population aging represents one of the most notable global demographic shifts of the 21st century, with estimates indicating that individuals aged 65 and over will constitute nearly 23% of the world population by 2054 (*1*). Accompanying this profound demographic transition is an unprecedented rise in age-related diseases, especially neurodegenerative disorders such as Alzheimer's disease (AD) (*2*, *3*). AD, the most common cause of dementia, manifests as a progressive neurodegenerative condition characterized by irreversible cognitive deterioration and currently lacks effective disease-modifying therapies. Despite extensive global efforts and approximately 200 clinical trials aimed at developing therapeutics for AD, nearly all have failed, with an alarmingly high failure rate approaching 99.6% (*4*, *5*). Although recently approved monoclonal antibodies, such as aducanumab and lecanemab, have shown some clinical efficacy, their therapeutic use is severely limited by clinical toxicity and safety concerns associated with long-term administration (*6*). Consequently, pharmacological interventions that can be safely and chronically administered to effectively prevent or halt disease progression remain virtually nonexistent.

Aging is consistently recognized as the primary risk factor for AD, offering critical insights into disease etiology and progression (*7*). Mechanistically, aging-related decline in neurogenesis, synaptic integrity, and neuronal function directly contributes to AD pathogenesis (*8*, *9*). At the cellular level, aging is characterized by cellular senescence, marked by irreversible cell cycle arrest, sustained DNA damage responses, and the senescence-associated secretory phenotype (SASP) (*10*). The SASP promotes chronic tissue inflammation, exacerbating aging-related pathologies, and substantial evidence has underscored neuroinflammation as a critical link between aging and AD (*3*, *11*, *12*). Furthermore, cumulative oxidative stress—including reactive oxygen species (ROS) accumulation, mitochondrial dysfunction, DNA damage, and epigenetic alterations—has emerged as another pivotal shared mechanism underlying both brain aging and AD pathology (*13*, *14*). Despite these insights, the precise relationship between aging and AD remains poorly defined. Specifically, it is unresolved whether aging directly initiates AD pathogenesis or whether AD pathology reciprocally accelerates aging processes, representing a fundamental puzzle in geroscience and neurodegeneration research (*15*, *16*). Given this complexity, unraveling the precise interplay between aging and AD has proven challenging. Thus, therapeutic strategies aimed at simultaneously targeting both aging mechanisms and AD pathogenesis hold immense clinical promise and may revolutionize the treatment landscape for age-related neurodegenerative diseases.

Sleep disorders, particularly insomnia, have emerged as an important bridge linking aging and AD (*17*, *18*). Insomnia affects over 16% of adults worldwide (more than 850 million people) and shows marked age-dependent prevalence (*19*). Accumulating evidence now supports that insomnia is not merely a comorbidity but a pathological accelerator of both aging and AD (*20*, *21*). Insomnia elevates neuronal activity, increasing amyloid-β (Aβ) production and aggregation while simultaneously inducing sympathetic hyperactivity that suppresses glymphatic clearance of Aβ, tau, and α-synuclein. Concomitantly, sleep-circadian disruption promotes oxidative stress,

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neuroinflammation, and synaptic homeostasis loss, collectively driving aging and neurodegeneration. Aging and neurodegeneration further exacerbate sleep deprivation, establishing a self-reinforcing neurodegeneration vicious circle (*22*). Hence, insomnia represents a modifiable risk factor and therapeutic entry point for AD and aging, underscoring the rationale for multitarget therapeutics that comodulate sleep, aging mechanisms, and AD pathology. However, despite this therapeutic rationale, conventional drug development pipelines remain poorly equipped to address the interconnected hallmarks of these conditions.

Although the substantial promise of multitargeted geroprotective therapies, conventional drug discovery pipelines struggle profoundly with the complexity and interconnectedness inherent to aging hallmarks, such as redox dyshomeostasis, chronic inflammation, and mitochondrial dysfunction (*23*). This complexity has led to protracted, costly development cycles and exceptionally high failure rates in clinical trials for aging-related disorders, including AD. Consequently, there is an urgent need to leverage transformative methodological innovations capable of efficiently pinpointing therapeutic candidates with genuine clinical potential. In this regard, artificial intelligence (AI), an emerging frontier integrating computer science, systems biology, and biomedical research, has demonstrated remarkable promise (*24*). By harnessing sophisticated computational frameworks—including machine learning, deep learning, and network-based analytics—to systematically mine and interpret vast genomic, transcriptomic, and pharmacological datasets, AI methodologies have revolutionized drug discovery processes. These techniques not only accelerate drug repurposing, molecular property prediction, and mechanistic deconvolution of therapeutic candidates but also mitigate the high costs and translational risks typically associated with traditional development approaches (*25*, *26*). Thus, AI-driven drug discovery emerges as an innovative and compelling pathway toward breakthroughs in the development of clinically effective therapeutics targeting the intertwined complexities of aging and AD.

To address the challenges and harness the immense therapeutic potential of dual-targeting interventions for aging and AD, we developed an AI-guided method termed the pathway and transcriptome-driven drug efficacy predictor (PTD-DEP). This model was specifically designed for the systematic identification and optimization of small-molecule candidates capable of targeting shared pathological pathways underlying aging and AD. Leveraging PTD-DEP, we identified melatonin (MLT), an endogenous hormone commonly used clinically as an anti-insomnia agent, as a promising candidate exhibiting dual antiaging and anti-AD therapeutic potential, which we subsequently validated through comprehensive in vitro and in vivo studies. Mechanistically, guided by proteolysis targeting chimera (PROTAC) technology combined with CB-Dock2 computational prediction, we uncovered p300 as a critical molecular target of MLT. Further integrative analyses using Cleavage Under Targets and Tagmentation (CUT&Tag), immunoprecipitation–mass spectrometry (IP-MS), single-cell RNA sequencing (scRNA-seq), and spatial transcriptomics, complemented by rigorous pharmacological validations, revealed that MLT specifically targets the p300/specificity protein 1 (SP1) transcriptional complex localized within super-enhancer regions. This targeted interaction potently drives transcriptional activation of brain and muscle arnt-like protein 1 (BMAL1), a pivotal regulator of circadian rhythm (*27*). Notably, the therapeutic efficacy of MLT was robustly confirmed across preclinical models of both AD and cellular senescence, underscoring its specific polypharmacological capacity to concurrently mitigate aging hallmarks and AD pathology.

Collectively, our findings not only provide a clinically relevant therapeutic candidate for AD prevention and treatment but also represent a notable theoretical advancement in geroscience by elucidating a novel molecular link between aging and AD. Moreover, our AI-driven methodological framework offers a transformative approach to overcoming traditional drug development barriers, opening new avenues for precise multitarget therapeutics and advancing translational strategies for complex aging-related neurological disorders.

RESULTS

AI-driven PTD-DEP for multitarget therapeutic prediction

We first constructed a deep learning–based framework, PTD-DEP, integrating multiomics network analysis with mechanistic deep learning to systematically identify and prioritize candidate compounds targeting aging and AD-related pathways. PTD-DEP features a dual-modality architecture comprising a biological pathway module and a transcriptomic profiling component. The pathway module uses machine learning to quantify compound-pathway interactions for high-throughput prioritization, while the transcriptomic component implements a multimodal deep architecture integrating autoencoders with graph convolutional networks (GCNs) to achieve prediction of pharmacological profile, enabling further therapeutic potential evaluation (Fig. 1A).

To construct the biological pathway prediction module of PTD-DEP, we curated small-molecule libraries from PubChem for training and established the prediction module comprising multiple single-task models for key pharmacological pathways including anti-AD, antiaging, anti-inflammation, antioxidation, and blood-brain barrier (BBB) permeability (Fig. 1B). For multidimensional therapeutic prediction, each dataset was randomly split at the compound level into training and held-out test sets in an 80:20 ratio. The training portion was further used for model development and hyperparameter optimization via fivefold GridSearchCV and RandomizedSearchCV in a unified, multialgorithm pipeline, with pathway-specific protocols ensuring algorithm-pathway compatibility (Fig. 1C and fig. S1A). The machine learning layer demonstrated robust predictive capacity across all channels (fig. S1, B to F). In detail, performance evaluated on the independent held-out test sets exceeded 0.9066 in accuracy, 0.9026 in precision, and 0.9188 in area under the receiver operating characteristic curve (ROC-AUC) (Fig. 1, C and D, and fig. S1A).

To develop the transcriptomic prediction component, we used the L1000 genomic resource as the primary training dataset (*28*, *29*). Single-cell sequencing data were leveraged to systematic characterize transcriptional signatures in the brain, and cell line profiles demonstrating transcriptomic homology with human brain cell types in L1000 were systematically curated as the training database. Through multitiered curation of the Tabula Sapiens (*30*) data involving batch-effect correction, dimensionality reduction, and systematic cell type classification, we revealed 61 molecularly defined cell types in 25 distinct tissues (fig. S1G). Using cosine similarity, Euclidean distance, and Pearson correlation analyses, we mapped the relationships between L1000 cell lines and human tissue-specific cell types, generating cross-system similarity networks. The L1000 pharmacogenomic profiles from HCT116, human embryonic kidney (HEK) 293T, JURKAT, and nasopharyngeal carcinoma (NPC) cell lines demonstrating transcriptomic homology to key neural populations

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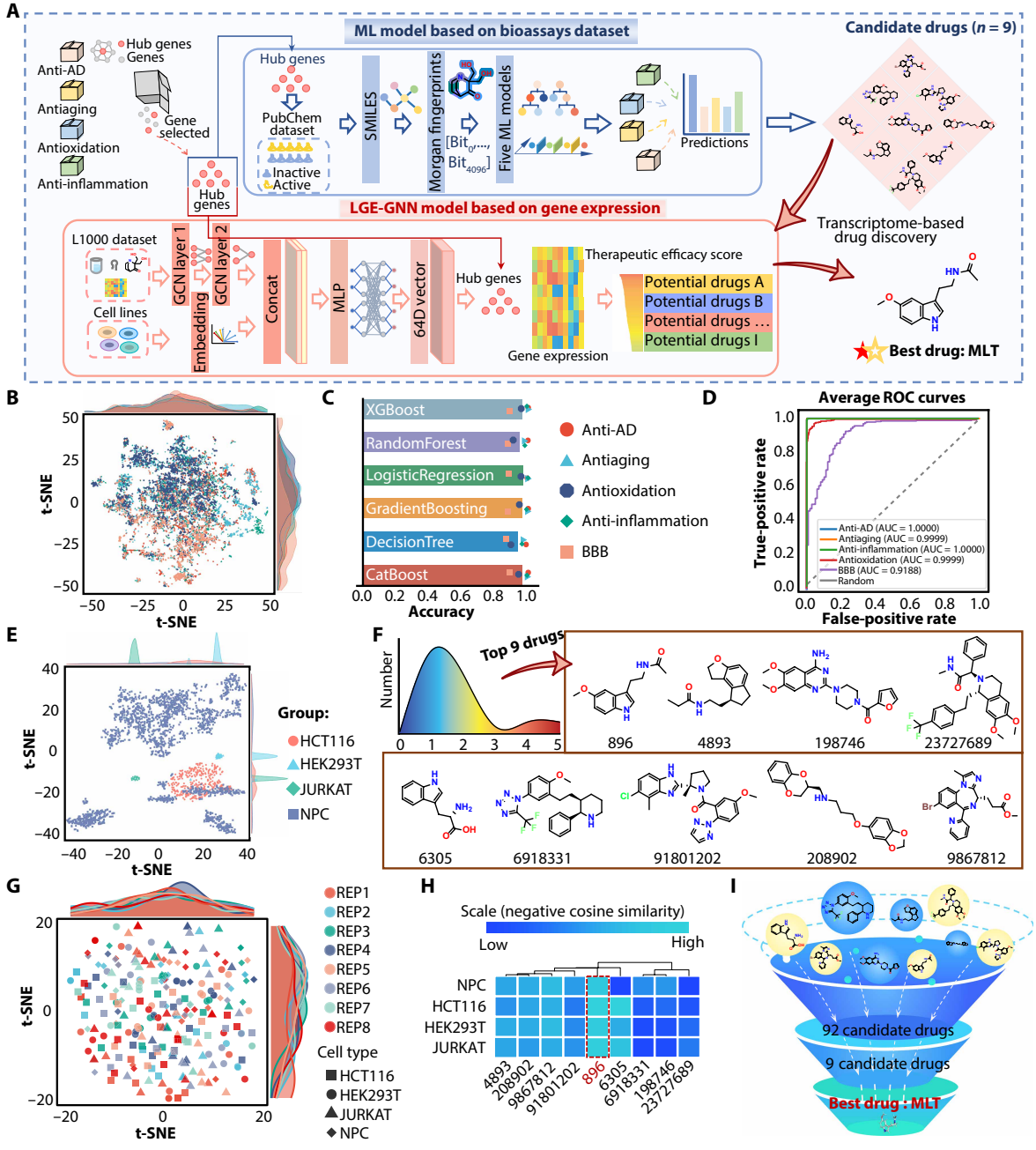


Fig. 1. AI-guided repurposing identifies MLT for aging and AD from anti-insomnia drugs. (**A**) Schematic overview of the PTD-DEP architecture and the workflow to screen candidate anti-AD drugs. LGE-GNN, Latent gene expression graph neural network. (**B**) *t*-distributed stochastic neighbor embedding (t-SNE) projection of the training compounds for the PTD-DEP biological pathway prediction module. (**C**) Bar plot showing held-out test set accuracies across multiple prediction tasks for the PTD-DEP biological pathway prediction module. (**D**) ROC curves illustrating the held-out test performance of the PTD-DEP biological pathway prediction module. (**E**) t-SNE projection of the molecular perturbation profiles used to train the transcriptomic prediction component of PTD-DEP. (**F**) Anti-insomnia drugs identified by the PTD-DEP biological pathway prediction module as potential anti-AD candidates. (**G**) t-SNE projection of molecular perturbation profiles for seven small molecules predicted by the PTD-DEP transcriptomic prediction component across four cell lines. (**H**) Heatmap depicting the negative cosine similarity between the predicted transcriptome and the disease-associated transcriptome. (**I**) Flowchart illustrating the comprehensive screening process for compounds sourced from anti-insomnia drugs using our PTD-DEP model.

(neurons, microglia, astrocytes, oligodendrocytes, and endothelial cells) were used as training database (Fig. 1E and fig. S1, H and I).

The transcriptomic prediction component architecture integrates autoencoders and GCN for transcriptome prediction. L1000 gene expression profiles were compressed into 64D latent vector using autoencoders, and then two layers of GCN were then used extract features from the molecular graphs generated by drug SMILES (Simplified Molecular Input Line Entry System). These molecular features were concatenated with cell line embeddings, and the combined representation was mapped to a predicted latent vector through a fully connected layer. The decoder subsequently reconstructed complete gene expression profiles from the predicted latent vector. The dataset was

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randomly split 80:20 into development set and held-out test set. The development set was further split 80:20 into training set and validation set (64%:16%:20% overall). Preprocessing parameters (scaling and clipping thresholds) were fit on the training set; model selection used validation R^2; the held-out test set was used only once for final evaluation. Upon completion of training, the optimized model achieved high accuracy in cross–cell line compound transcriptome prediction (R^2 on held-out test set = 0.7799) significantly exceeding that of linear regression (R^2 on held-out test set = 0.2563). By synergistically combining biological pathway analysis with transcriptomic profiling, the dual-modal PTD-DEP systematically leverages multiomics repositories to enable quantitative prediction of drug candidates' multitarget regulatory profiles, including both AD therapeutic efficacy and senescence-intervention capacity, with potential to accelerate drug development.

AI-guided repurposing identifies MLT for aging and AD from anti-insomnia drugs

To evaluate the practical utility of the validated PTD-DEP model in therapeutic discovery, we applied our computational screening pipeline specifically aimed at repurposing clinically approved anti-insomnia drugs to identify novel anti-AD therapeutic candidates with neuroprotective potential. Accumulating evidence suggests that sleep disturbances contribute to cognitive decline and increased AD risk (*31*). Mendelian randomization analysis further indicated that insomnia is a risk factor for AD (fig. S2, A to C). Therefore, after constructing the PTD-DEP framework, we aimed to identify potential anti-AD agents among existing anti-insomnia drugs, with the goal of drug repurposing and expanding therapeutic applications. Using the machine learning layer of PTD-DEP, nine compounds with potential anti-AD efficacy were identified from 92 anti-insomnia drugs (Fig. 1F and data S1). The transcriptomic profiles of these candidates were then predicted in four different cell lines based on the transcriptomic prediction component of PTD-DEP (Fig. 1G). Cosine similarity analysis with disease-state transcriptomic data ultimately singled out MLT, a neurohormone secreted by the pineal gland, which ranked first in the promising anti-AD candidate list (Fig. 1, H and I, and data S2). Moreover, cosine similarity analysis of L1000 expression profiles from 2438 compounds indicates that MLT has superior therapeutic potential for AD (data S2). Thus, this integrative AI-driven pipeline highlights MLT as a promising multitarget therapeutic candidate for AD, demonstrating the substantial translational potential of combining Mendelian randomization with advanced machine learning for rapidly identifying safe and effective therapies for complex neurological disorders.

Clinical analysis of 161 human samples reveals reduced serum MLT levels correlated with aging

Numerous studies have demonstrated that MLT extensive biological activities, including anti-inflammatory, antioxidant, and antiamyloidogenic pathway properties (*32*, *33*), suggesting its potential protective role against aging-related neurodegenerative disorders. To clinically validate the relevance of MLT in aging, we conducted a rigorous cross-sectional analysis involving 161 clinical serum samples, collected from human individuals across a wide age spectrum (Fig. 2A). Considering the intrinsic difficulty of detecting trace MLT concentrations in serum, we adopted a standard measurement technique using liquid chromatography–tandem MS (LC-MS/MS) methodology operated in multiple reaction monitoring (MRM) mode to ensure accurate quantification (Fig. 2B). Our quantitative analyses revealed a significant negative correlation between serum MLT levels and age ($r = -0.289$, $P < 0.001$), indicating that serum MLT progressively decreases with advancing age (Fig. 2C). To further verify this relationship, we stratified individuals into younger (<60 years) and elderly (≥60 years) groups. The elderly group demonstrated significantly lower serum MLT levels compared to the younger cohort, strongly affirming age-dependent MLT depletion (Fig. 2D). These clinical findings from a substantial cohort robustly establish that declining serum MLT concentrations correlate with aging, suggesting that diminished serum MLT could represent a critical risk factor or biomarker associated with aging and susceptibility to neurodegenerative diseases.

MLT supplementation alleviates cognitive impairment and pathological features of AD in Aβ precursor protein/presenilin-1 mice

To validate the potential of treating AD by MLT administration, we first performed in vitro experiments using human neuroblastoma SH-SY5Y cells overexpressing the Swedish mutant form of human Aβ precursor protein (APPswe cells). MLT significantly down-regulated APP, the cytokine interleukin-1β (IL-1β), and monocyte chemoattractant protein 1 (*MCP1*) (Fig. 3, A to C), indicating its potential to ameliorate AD-related pathological markers and neuroinflammatory responses in vitro. Next, to assess the therapeutic efficacy of MLT in ameliorating cognitive and memory dysfunction associated with AD, we administered the MLT supplement with water to 6-month-old APP/presenilin-1 (PS1) AD model mice (Fig. 3D). The Morris water maze (MWM) analysis was conducted to examine spatial learning and memory abilities. In the training stage of the MWM test, the APP/PS1 control mice spent more time to find the platform on training day 5 compared with wild-type (WT) mice, while MLT treatment significantly shortened the escape latency time on the 4th and 5th days (Fig. 3E). During testing, APP/PS1 mice exhibited impaired spatial navigation relative to WT controls, with significantly prolonged escape latency and increased pathlength to platform location, indicative of cognitive deficits (Fig. 3, F to H). MLT treatment ameliorated all these behavioral indices, demonstrating therapeutic efficacy against AD-associated cognitive impairment.

Furthermore, immunostaining with thioflavin S showed the reduction of Aβ cores in hippocampus and cortical tissues of MLT-treated APP/PS1 mice relative to vehicle group (Fig. 3I). Neuroinflammation, a key role in AD progression, increases the levels of proinflammatory cytokines in the central nervous system, contributing to cognitive decline (*34*). Subsequent quantitative polymerase chain reaction (qPCR) analysis indicated that MLT administration decreased levels of proinflammatory cytokines *Il1b*, *Il6*, and *Tnf* (Fig. 3J). In addition, MLT therapeutically attenuated cortical ROS accumulation in APP/PS1 mice, as evidenced by immunofluorescence quantification, concomitant with oxidative stress mitigation (Fig. 3K). Synaptic impairment and dysfunction are key pathological characteristics of AD, closely connected with the deterioration of cognitive function (*35*). Consequently, the protein levels of postsynaptic density protein 95 (PSD95; encoded by *Dlg4*) and synaptophysin (SYP) were up-regulated after MLT treatment in APP/PS1 mice (Fig. 3L). MLT administration significantly enhanced synaptic plasticity markers (*Dlg4*, *Syp*, *Gria2*, and *Camk2a*) in APP/PS1 mice, demonstrating that MLT may benefit AD by modulating synaptic function (Fig. 3M).

To further elucidate the role of MLT in AD pathogenesis, we conducted a bulk RNA-seq on brain tissues obtained from APP/PS1 mice with and without MLT administration, revealing 2670 differentially

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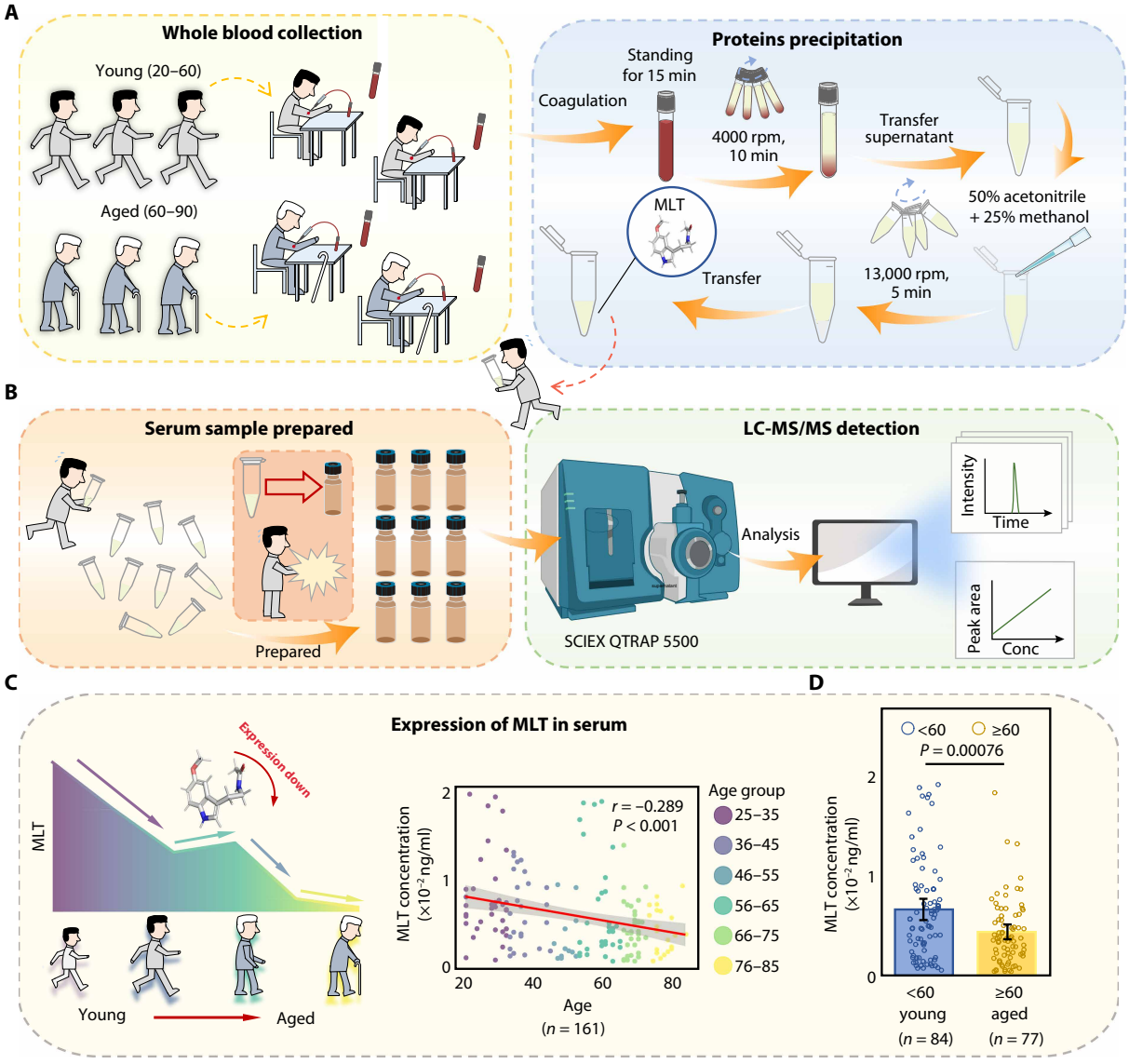


Fig. 2. Clinical analysis of 161 human samples reveals reduced serum MLT levels correlated with aging. (**A**) Schematic detailing the standard procedure for sample collection. (**B**) Flowchart illustrating the steps involved in LC-MS/MS analysis. (**C**) Correlation analysis between MLT concentration in plasma and ages ($n = 161$). (**D**) Comparative MLT concentrations in plasma samples obtained from donors of different ages ($n = 84$ in the young group and 77 in the aged group). Each dot represents an individual donor.

expressed genes (DEGs) (1172 up-regulated genes and 1498 down-regulated genes; Fig. 3, N and O, and data S3). Gene Ontology (GO) analysis of DEGs following confirmed the neuroprotective role of MLT, as demonstrated by the DEGs enriched in inflammatory response, neuron-to-neuron synapse, and antioxidant activity (Fig. 3P). Together, these comprehensive findings highlight that MLT supplementation robustly mitigates AD-associated cognitive impairment and neuropathological changes by simultaneously attenuating neuroinflammation, reducing oxidative stress, alleviating amyloid pathology, and promoting synaptic recovery. Our results thus strongly support the therapeutic potential of MLT as a promising anti-AD agent, either alone or in combination therapies, paving the way for future translational research.

MLT ameliorates cellular senescence in mesenchymal stem cells via modulation of senescence-associated pathways

To further elucidate the antiaging potential of MLT, we used human mesenchymal stem cells (MSCs), which have recently emerged as an attractive and clinically relevant model for investigating cellular aging. Late-passage MSCs typically exhibit prominent senescent characteristics, including elevated oxidative stress, enhanced SASP, mitochondrial dysfunction, and irreversible cell cycle arrest, thus providing an ideal experimental system for evaluating therapeutic interventions targeting senescence (*36*).

As expectedly, MLT treatment significantly reduced senescence-associated β-galactosidase (SA-β-Gal)–positive cells and attenuated ROS accumulation in aged MSCs (Fig. 4, A and B). MLT suppressed

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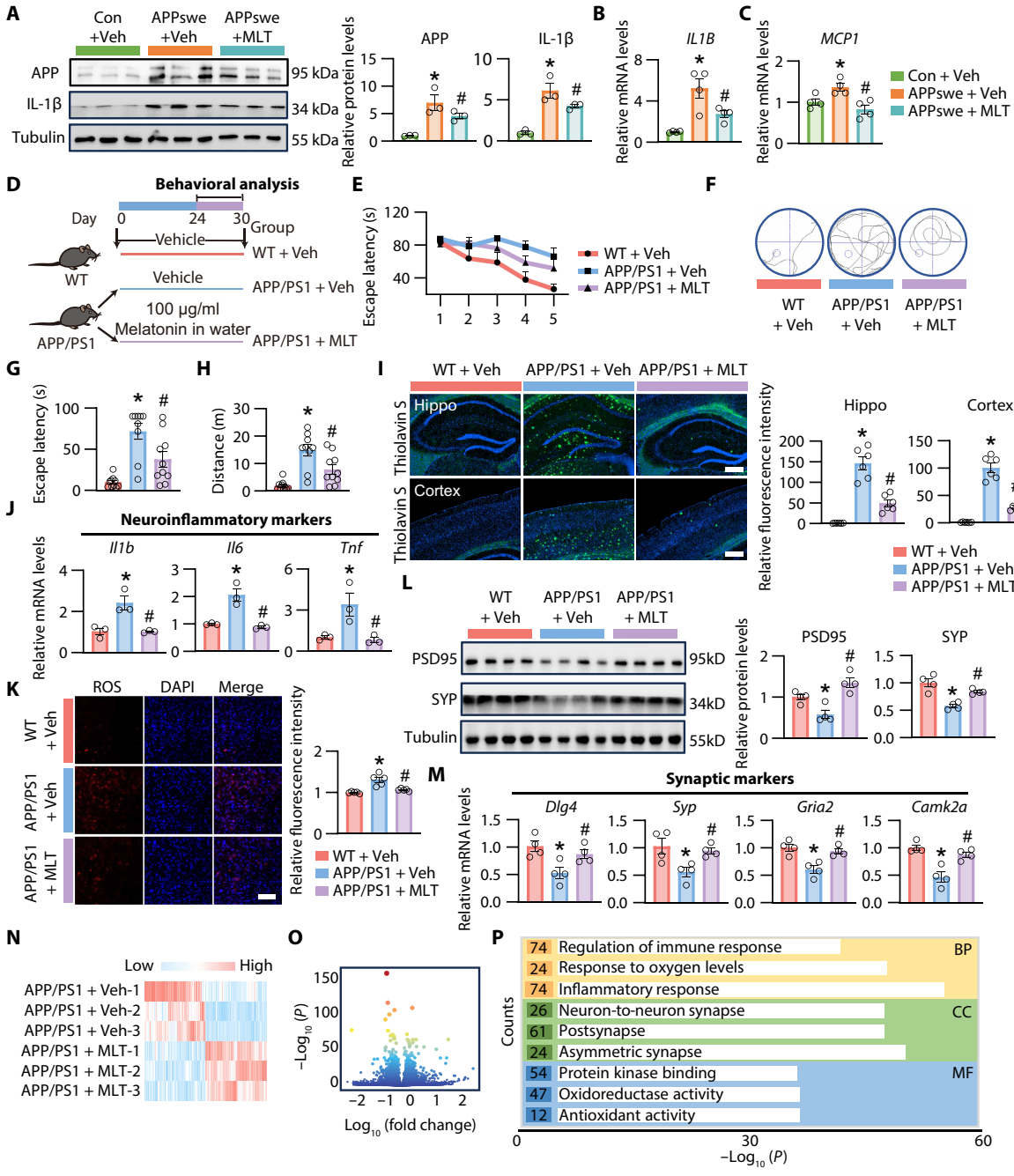


Fig. 3. MLT supplementation alleviates cognitive impairment and pathological features of AD in vitro and in vivo. (**A**) Western blot assays and quantification analysis show the relative protein level of APP and IL-1β ($n = 3$ per group). (**B** and **C**) Relative mRNA levels of *IL1B* and *MCP1* ($n = 4$ per group). Veh, vehicle. (**D**) Schematic diagram showing treatment and behavioral assessment. (**E** to **H**) Assessment of the cognitive function by the MWM test. Escape latency during training days (E), representative video tracking images in final test (F), escape latency in final test (G), and distance to the target platform in final test (H) ($n = 9$ in APP/PS1+veh and 10 in WT and MLT groups). (**I**) Thioflavin S staining and quantification analysis of hippocampal and cortex ($n = 6$ per group). Scale bars, 200 μm. (**J**) Relative mRNA levels of neuroinflammatory markers (*Il1b*, *Il6*, and *Tnf*) ($n = 3$ per group). (**K**) Representative fluorescence micrographs and statistical analysis of dihydroethidium (DHE) staining in the cortex ($n = 5$ per group). Scale bar, 100 μm. (**L**) Western blot assays and quantification analysis show the relative protein level of postsynaptic density protein 95 (PSD95) and synaptophysin (SYP) ($n = 4$ per group). (**M**) Relative mRNA levels of synaptic markers (*Dlg4*, *Syp*, *Gria2*, *and Camk2a*) ($n = 4$ per group). (**N**) Heatmap depicting global transcriptomic alterations in APP/PS1 with/without MLT treatment. (**O**) Volcano plot showing DEGs following MLT administration. (**P**) Lollipop plot summarizing biological pathways significantly enriched among the genes highlighted in (O). BP, biological process; CC, cellular component; MF, molecular function. Data are shown as means ± SEM. *$P < 0.05$ versus control group or WT group; #$P < 0.05$ versus APPswe group or APP/PS1 group.

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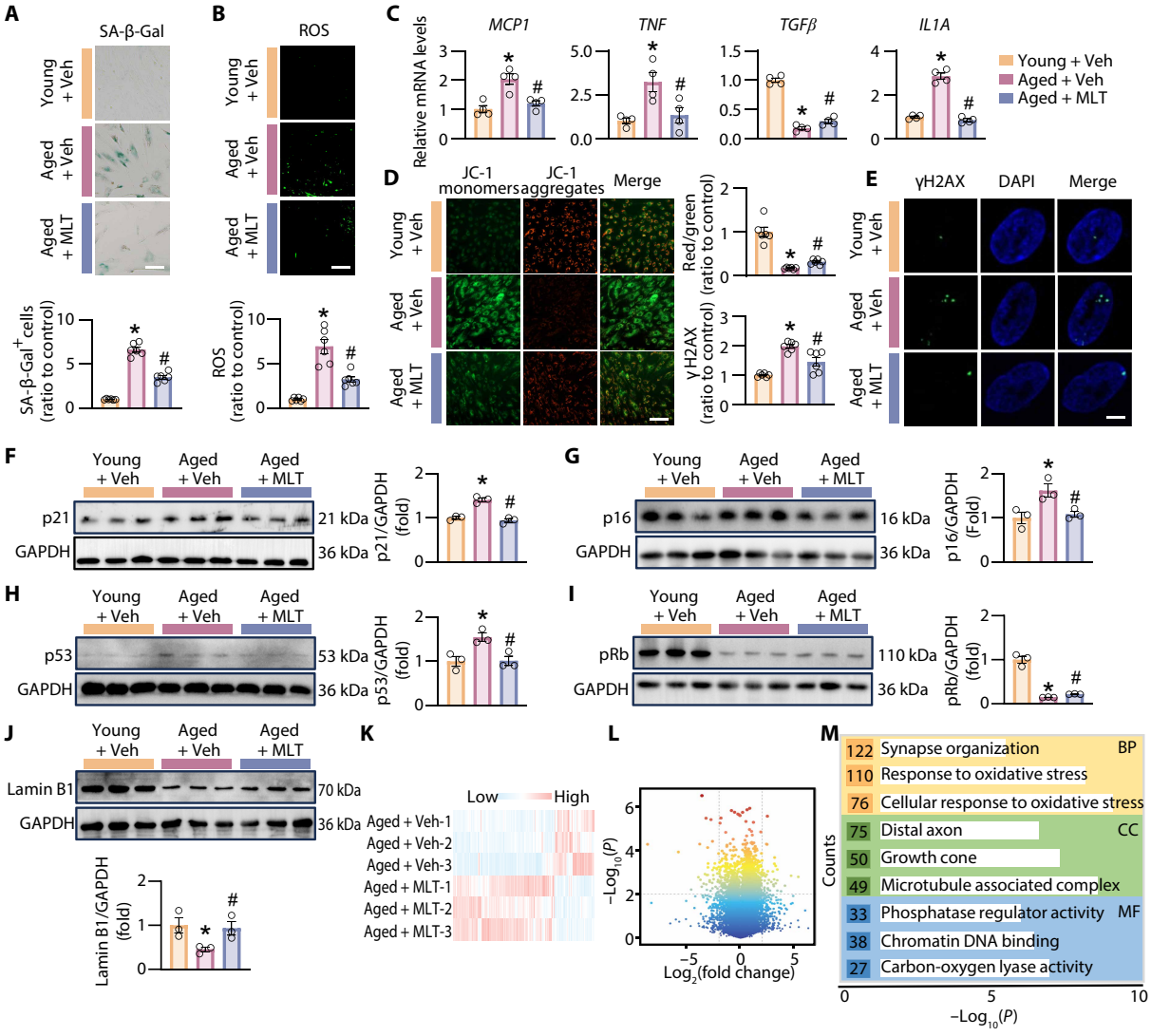


Fig. 4. MLT ameliorates cellular senescence in MSCs via modulation of senescence-associated pathways. (**A** and **B**) Representative immunofluorescent images and quantification analysis of SA-β-Gal (A) and ROS (B) ($n = 6$ per group) in MSCs. Scale bars, 100 μm. (**C**) Relative mRNA levels of SASPs (*MCP1*, *TNF*, *TGFβ*, and *IL1A*) ($n = 4$ per group). (**D**) Representative immunofluorescent images and quantification analysis of JC-1 staining to detect MMP ($n = 6$ per group). Scale bar, 100 μm. (**E**) Representative immunofluorescent images and quantification analysis of γH2AX foci ($n = 6$ per group). Scale bar, 10 μm. (**F** to **J**) Western blot assays and quantification analysis show the relative protein level of p21 (F), p16 (G), p53 (H), pRb (I), and Lamin B1 (J) ($n = 3$ per group). GAPDH, glyceraldehyde-3-phosphate dehydrogenase. (**K**) Heatmap of gene expression changes pre– and post–MLT treatment in aged MSCs. (**L**) Volcano plot of DEGs after MLT exposure on aged MSCs. (**M**) Lollipop plot of pathways significantly enriched among the genes in (L). Data are shown as means ± SEM. *$P < 0.05$ versus young group; #$P < 0.05$ versus aged group.

SASP components (*MCP1*, *TNF*, and *IL1A*) while up-regulating *TGF*β expression in senescent cells (Fig. 4C). One of the hallmarks of aging is the occurrence of mitochondrial dysfunction (*37*). Mitochondrial dysfunction in aged MSCs was demonstrated by diminished mitochondrial membrane potential (MMP), evidenced by an increased JC-1 monomer (green)/aggregate (red) fluorescence ratio that was rescued with MLT treatment (Fig. 4D). MLT further attenuated H2A histone family member X (H2AX) phosphorylation, a hallmark of DNA damage stress in senescence (Fig. 4E). Senescent cells exhibit hallmark up-regulation of cyclin-dependent kinase inhibitors $p21^{CIP1}$ and $p16^{INK4A}$, contributing to cell cycle arrest (*38*). The persistent DNA damage response in aging activates tumor suppressor p53, which mediates downstream senescence pathways (*39*). Notably, MLT significantly attenuated age-associated elevation of p21, p16, and p53 protein levels in aged MSCs, indicating its potential antiaging mechanism through senescence pathway modulation (Fig. 4, F to H). The up-regulation of these proteins results in hypophosphorylated Rb, a senescence-related signaling outcome notably counteracted by pharmacological intervention with MLT (Fig. 4I). Blocks of mitosis in senescent cells lead loss of expression of the nuclear lamina protein Lamin B1 (*40*). Senescent MSCs exhibit marked down-regulation of Lamin B1, which is pharmacologically restored by MLT intervention (Fig. 4J). Pharmacologically, MLT administration ameliorates senescence-associated hallmark alterations in aged MSCs.

Complementarily, to investigate mechanistic contributions of MLT in aging physiological and pathological processes, we subjected

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MLT-treated/untreated MSCs to RNA-seq, identifying 5236 DEGs (3982 up-regulated and 1254 down-regulated; Fig. 4, K and L, and data S4). GO analysis conducted on MLT-responsive DEGs revealed antisenescence mechanisms through significant enrichment in oxidative stress response, chromatin DNA binding, and synaptic plasticity pathways (Fig. 4M).

Collectively, our findings highlight that MLT exhibits potent geroprotective efficacy in MSCs by comprehensively targeting key senescence-associated pathways, such as oxidative stress, mitochondrial function, DNA damage response, SASP regulation, and cell cycle control. Given the rising therapeutic interest in MSCs for regenerative medicine, these results underscore the translational potential of MLT as a promising intervention against aging-related degenerative diseases, including AD.

PROTAC-based chemoproteomic identification of p300 as a target of MLT

Given accumulating evidence of MLT's neuroprotective efficacy, we next investigate its mechanistic underpinnings in modulating AD progression and age-related pathogenesis. PROTAC technology represents a bifunctional modality that hijacks the endogenous ubiquitin-proteasome system for targeted protein degradation (*41*). PROTAC-mediated proteolysis operates through transient target engagement, empowering chemoproteomic strategies to elucidate pharmacological targets (*42*).

To deconvolve MLT's mechanistic targets, we implemented PROTAC-enabled chemoproteomic screening, establishing target engagement landscapes for MLT-mediated neuroprotection. (Fig. 5A). The PROTAC development initiative commenced with rational design of an MLT-based heterobifunctional degrader (MLT-PROTAC, designated M-PROTAC). The synthetic route features conjugation of MLT to thalidomide through polyethylene glycol–based linker (fig. S3).

To further explore the potential targets degraded by M-PROTAC (Fig. 5B), we applied a quantitative proteomics approach to monitoring the protein fold changes at the cellular level and loaded the MLT as a negative control to exclude the undesired substrates. The proteomics proteins of Neuro-2a (N_2A) cells after 24 hours of treatment with 10 μM M-PROTAC or MLT were compared. Quantitative proteomic profiling identified 84 distinct targets, with 32 demonstrating significant down-regulation ($P < 0.05$; Fig. 5C and data S5), including the top 10 candidate proteins exhibiting maximal degradation efficacy (Fig. 5D). To mechanistically elucidate MLT's validated pharmacological targets, we performed the computational interrogation via CB-Dock2 to predict interactions between proteome-derived candidates and MLT (*43*), with EP300, histone acetyltransferase p300, emerging as the top-ranked candidate exhibiting superior binding affinity (Fig. 5, E and F). Subsequent data demonstrated that p300 was significantly degraded by M-PROTAC in dose- and time-dependent manners (Fig. 5, G and H). Cellular thermal shift assay (CETSA) validated direct MLT-p300 engagement, with a difference in melting temperature (ΔTm) shift demonstrating ligand-induced protein stabilization (Fig. 5I). Together, these data demonstrated that p300 is verified to be the underlying molecular target of MLT.

As a transcriptional coactivator, p300 orchestrates context-dependent gene regulation and plays a pivotal role in cellular proliferation, apoptosis, and embryogenesis (*44*). To investigate p300 chromatin occupancy dynamics post–MLT engagement, we used CUT&Tag for genome-wide profiling, revealing that MLT-induced augmentation of p300 chromatin occupancy, demonstrating MLT-mediated p300's transcriptional regulatory potentiation (Fig. 5J). Integrated of MACS2-identified differential peak analysis and GO enrichment analysis demonstrated MLT-driven enhancement of p300-mediated transcriptional modulation in neuroinflammatory response and neuron-protective pathways (Fig. 5K). Next, pharmacological perturbation of p300 using the selective inhibitor C646 was implemented to delineate its functional necessity in MLT's protective mechanisms. C646 pretreatment abrogated MLT-mediated suppression of APP expression in APPswe cells (Fig. 5L), while concomitantly blocking MLT's attenuation of proinflammatory mediators *IL1B* and *MCP1* (Fig. 5, M and N). Furthermore, C646 pharmacologically antagonized MLT-mediated senostatic effects in MSCs, reversing SA-β-Gal activity attenuation (Fig. 5O) and SASP (including *MCP1*, *TNF*, and *IL1A*) suppression (Fig. 5P). Notably, the p300 inhibitor countered MLT-mediated inhibition of senescence-associated marker p21 while attenuating MLT-induced up-regulation of Lamin B1 (Fig. 5, Q and R). In addition, C646 elevated H2AX phosphorylation in aged MSCs relative to MLT treatment group (Fig. 5S). Together, these results demonstrate that p300 is a critical molecular target mediating MLT's protective effects across both neurodegenerative and aging-related cellular contexts. The identification of p300 via PROTAC-based chemoproteomics not only elucidates a previously unrecognized mechanism of MLT action but also provides a compelling pharmacological rationale for its development as a multitarget intervention for age-associated disorders such as AD.

MLT modulates p300/SP1-associated super-enhancer landscape to regulate BMAL1 transcription

To delineate p300's mechanistic role in MLT-conferred neuroprotection, we performed IP-MS–based interactome profiling. IP-MS profiling identified 211 high-confidence p300 interactors, with 193 demonstrating cross-species conservation between human and mouse proteomes (fig. S4A and data S6). To elucidate the functional implications of p300 interactome, we implemented the protein-protein interaction prediction via microenvironment-aware protein embedding (MAPE-PPI) (*45*) computational framework for systematic prediction of protein interaction networks. Specifically, we leveraged AlphaFold3 for p300's structural prediction and applied the method provided by MAPE-PPI to construct a residue-level heterogeneous protein graph for p300. Next, we encoded this graph with a pretrained masked codebook and predicted the interaction types of p300 and its interactors (Fig. 6A). MAPE-PPI analysis identified 139 high-confidence p300 interactors within the PPI network, including 42 proteins with functionally active interfaces validated by binding data (data S7). Subsequent STRING database analysis combined with cytoHubba screening identified 15 hub proteins for structural characterization (Fig. 6A and fig. S4B). AlphaFold3-predicted p300-compound complexes were subjected to Rosetta binding energy assessment (dG_cross/dSASA×100, dG_separated, and dG_REU), with SP1 emerging as the top-ranked p300 binder through consensus scoring (Fig. 6B and data S8). Co-IP assays further confirmed p300-SP1 physical interaction (Fig. 6C). Collectively, MAPE-PPI predictions and experimental validation demonstrate that p300 interacts with and activates SP1, forming a transcriptional complex that modulates downstream gene expression.

To elucidate p300's functional role post–SP1-binding activation, we leveraged the DeepD2V to predict cobound DNA motifs (*46*). Using cross–cell line chromatin IP sequencing (ChIP-seq) profiles of p300 and SP1curated from ReMap2022 (*47*), we constructed the

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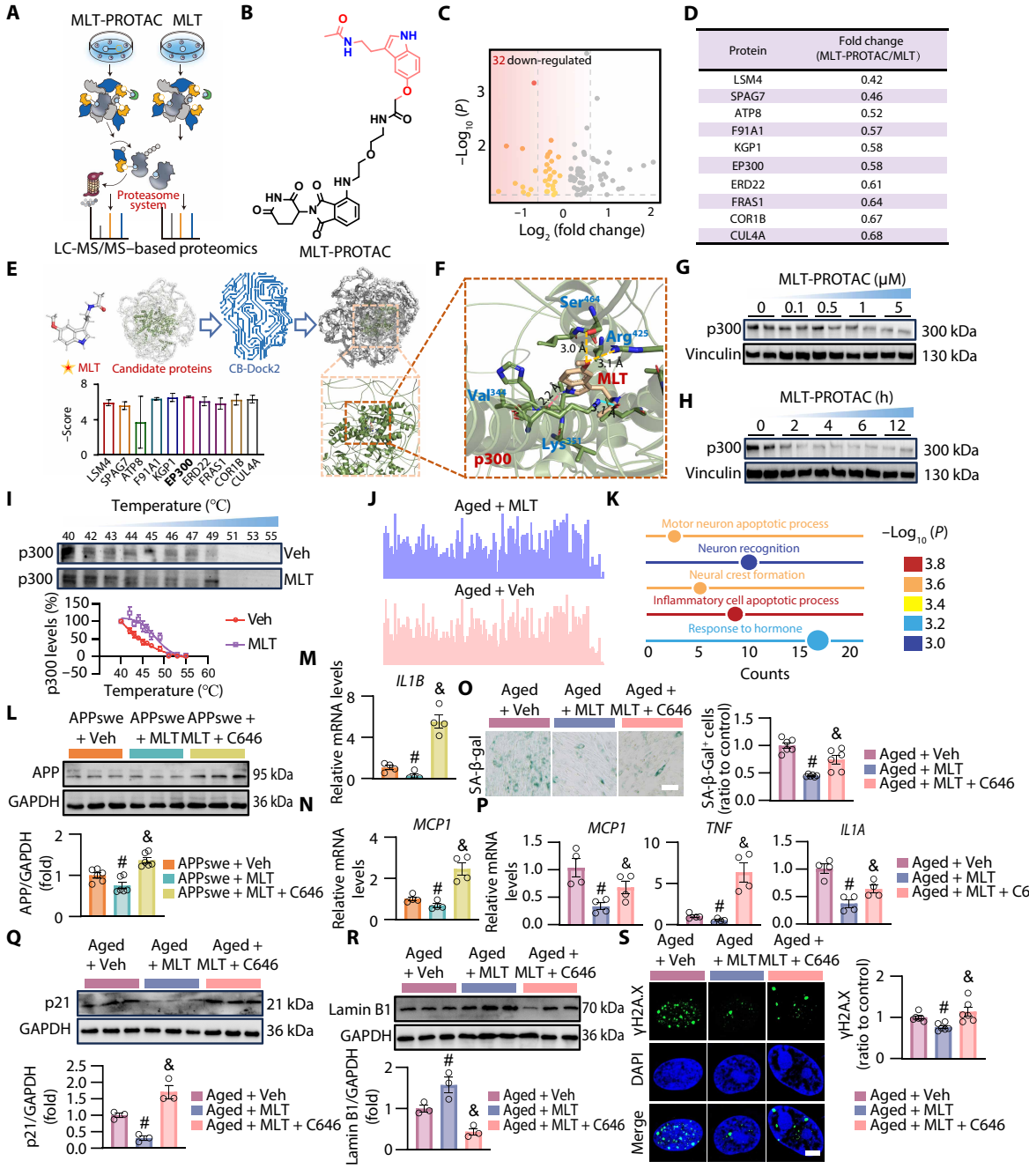


Fig. 5. PROTAC-based chemoproteomic identification of p300 as a target of MLT. (**A**) LC-MS/MS–based approach used to assess the proteome changes in N_2A after 24-hour treatment of M-PROTAC. (**B**) Chemical structures of M-PROTAC. (**C**) Volcano plot showing differential expressed proteins after M-PROTAC treatment (M-PROTAC/control). (**D**) Top 10 proteins with the most degradation after M-PROTAC treatment. LSM4, Sm-like protein 4 homolog; SPAG7, Sperm Associated Antigen 7; ATP8, Mitochondrially encoded ATP synthase membrane subunit 8; F91A1, Family with sequence similarity 91 member A1; KGP1, Protein kinase cGMP-dependent 1; ERD22, KDEL endoplasmic reticulum protein retention receptor 2; FRAS1, Fraser extracellular matrix complex subunit 1; COR1B, Coronin 1B; CUL4A, Cullin 4A. (**E**) Bar plot of CB-Dock2 predicted binding affinities between MLT and 10 candidate proteins. (**F**) Predicted binding orientation of MLT in the p300 active site (CB-Dock2). (**G** and **H**) Western blot assays and quantification analysis show the (G) concentration- and (H) time-dependent degradation of p300 protein in N_2A cells ($n = 4$ per group). h, hours. (**I**) CETSA to detect the stability of p300. (**J**) Integrative Genomics Viewer (IGV) snapshots illustrating the redistribution of p300 binding sites across the genome following MLT treatment. (**K**) Lollipop plot showing the biological pathways that are significantly enriched among genes regulated by p300. (**L**) Western blot assays and quantification analysis show the relative protein level of APP ($n = 6$ per group). (**M** and **N**) Relative mRNA levels of *IL1B* (M) and *MCP1* (N) ($n = 4$ per group). (**O**) Representative immunofluorescent images and quantification analysis of SA-β-Gal in MSCs ($n = 6$ per group). Scale bar, 100 μm. (**P**) Relative mRNA levels of SASPs (*MCP1*, *TNF*, and *IL1A*) ($n = 4$ per group). (**Q** and **R**) Western blot assays and quantification analysis show the relative protein level of p21 (Q) and Lamin B1 (R) ($n = 3$ per group). (**S**) Representative immunofluorescent images and quantification analysis of γH2AX foci ($n = 6$ per group). Scale bar, 10 μm. Data are shown as means ± SEM. #$P < 0.05$ versus APPswe group or aged group; &$P < 0.05$ versus APPswe + MLT group or aged + MLT group.

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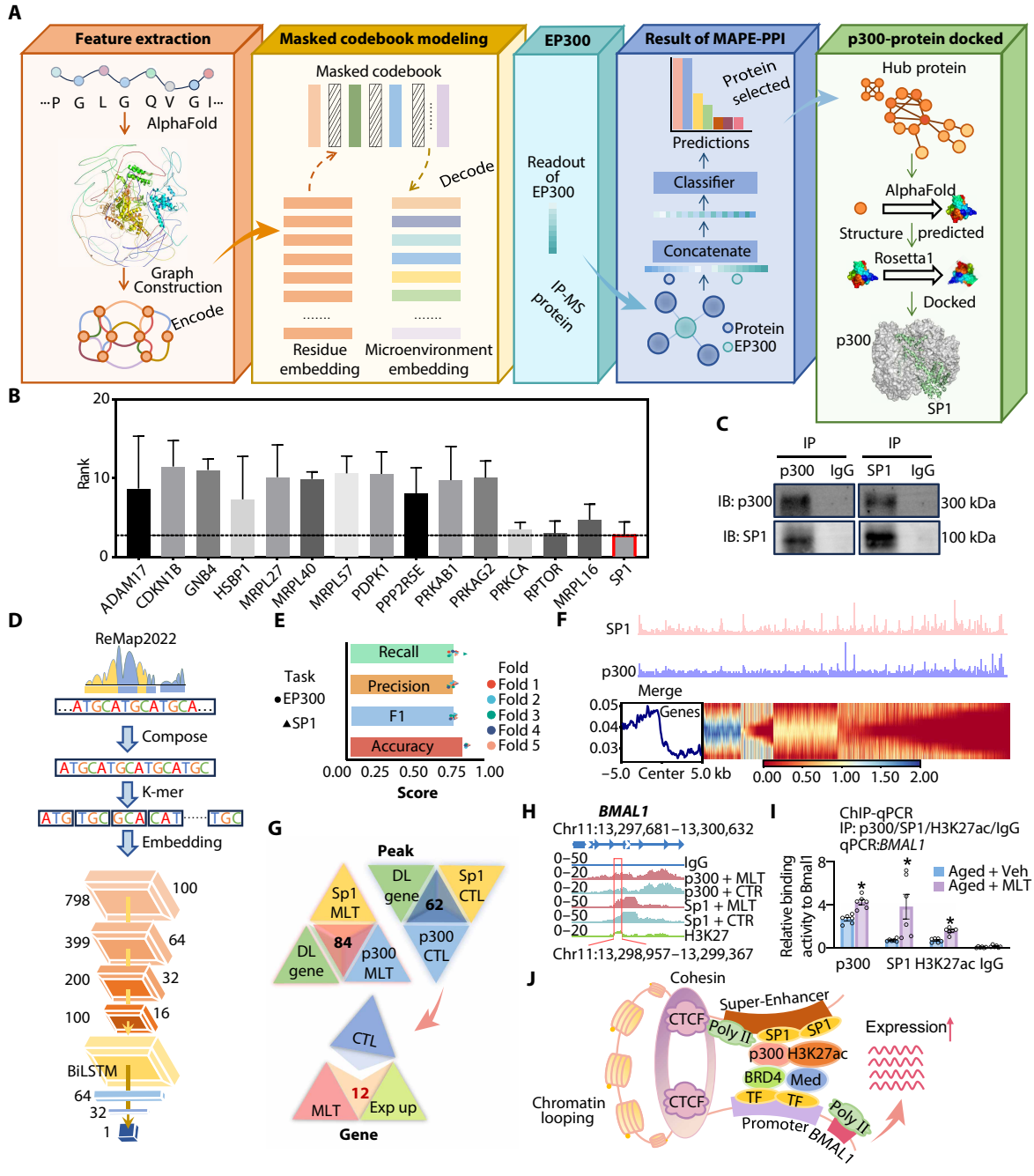


Fig. 6. MLT modulates p300/SP1-associated super-enhancer landscape to regulate BMAL1 transcription. (**A**) Workflow combining MAPE-PPI, AlphaFold3, and Rosetta to prioritize proteins with high binding affinity to p300. (**B**) Composite ranking of Rosetta scores for p300 complexes with various proteins. ADAM17, A disintegrin and metalloproteinase 17; CDKN1B, Cyclin-dependent kinase inhibitor 1B; GNB4, G protein subunit beta 4; HSBP, Heat shock factor binding protein; MRPL27, Mitochondrial ribosomal protein L27; PDPK1, 3-phosphoinositide-dependent protein kinase 1; PPP2R5E, Protein phosphatase 2 regulatory Subunit B′Epsilon; PRKAB1, Protein kinase AMP-activated non-catalytic subunit beta 1; RPTOR, Regulatory associated protein of mTOR, Complex 1. (**C**) Co-IP confirming the interaction between p300 and SP1. IB, immunoblotting. (**D**) Schematic of the DeepD2V pipeline used to predict DNA fragments cobound by p300 and SP1. (**E**) Bar plot showing the performance of the DeepD2V in predicting the DNA sequence bound to the p300 and SP1. (**F**) IGV tracks and heatmap showing genome-wide cobinding of p300 and SP1 predicted by DeepD2V. (**G**) Venn diagram illustrating genes that are up-regulated after MLT treatment and regulated by SP1/p300. DL, Deep learning, referred to DeepD2V model. (**H**) IGV visualization reveals colocalization of SP1 and p300 at a *BMAL1* regulatory locus marked by strong H3K27ac enrichment. (**I**) ChIP-qPCR assay demonstrated that p300/SP1/H3K27ac binding to *Bmal1* SE regions was enhanced after MLT administration ($n = 6$ per group). (**J**) Schematic diagram representing the BMAL1 transcription regulated by p300/SP1 complex via SE. CTCF, CCCTC-binding factor; BRD4, Bromodomain protein 4; TF, transcription factor. Data are shown as means ± SEM. $^*P < 0.05$ versus aged group.

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DeepD2V training dataset. Posttraining, DeepD2V demonstrated high prediction accuracy for DNA binding to SP1 (0.9336) and p300 (0.9156) (Fig. 6, D and E, and fig. S4, C and D). Subsequently, the human genome was partitioned into 1,098,591 genomic segments for independent p300 and SP1 binding propensity predictions. Independent prediction revealed 3653 loci with p300-SP1 co-occupancy (Fig. 6F and data S9), showing significant GO enrichment regulation of nervous system, learning or memory, and forebrain development, suggesting functional modulation in these pathways (fig. S4E). To investigate MLT-induced regulatory dynamics of p300/SP1 complexes, we conducted CUT&Tag experiments (fig. S4, F to J). Integrated analysis of DeepD2V predictions, CUT&Tag data (p300/SP1), and RNA-seq profiles pre–/post–MLT administration revealed cobound genes exhibiting enhanced transcriptional activity with increased p300/SP1 occupancy. Systematic analysis identified 12 MLT-regulated downstream targets of the p300/SP1 complex (Fig. 6G).

As a core epigenetic modulator, p300 functions not only as a transcriptional coactivator but also as an essential super-enhancer (SE) constituent, with its genomic occupancy serving as a hallmark of SE activation (*48*). Given MLT's modulatory role in p300 activity, we propose that MLT orchestrates transcriptional regulation through p300-mediated SE formation. Leveraging the established association between SEs and histone H3 lysine-27 acetylation (H3K27ac) enrichment, we performed H3K27ac CUT&Tag sequencing to characterize active chromatin domains, revealing pronounced acetylation landscapes (fig. S4, K and L). Analysis revealed elevated H3K27ac levels in 10 p300/SP1 cobound genes (fig. S4M and data S10). BMAL1, a circadian rhythm–regulating transcription factor linked to AD and aging (*49*, *50*), displayed MLT-enhanced p300/SP1 co-occupancy at its promoter with concomitant H3K27 hyperacetylation, corroborated by ChIP-qPCR validation (Fig. 6, H and I). This MLT-driven chromatin remodeling establishes a SE configuration that mechanistically up-regulates BMAL1 expression (Fig. 6J). Overall, our study demonstrates that MLT activates the p300/SP1 transcriptional complex and promotes super-enhancer formation to drive BMAL1 expression. This epigenomic remodeling mechanism reveals a central axis linking MLT to circadian rhythm regulation, neuroinflammation control, and aging-associated transcriptional plasticity, underscoring its potential as a regulator of super-enhancer–mediated gene programs in neurodegeneration and senescence.

BMAL1 functions as a central effector of MLT-mediated neuroprotection and antisenescence activity

Collective evidence demonstrates that MLT facilitates p300/SP1 complex formation and subsequently enhances BMAL1 transcriptional activation. To evaluate MLT's therapeutic potential, we interrogated BMAL1/p300/SP1 expression profiles by integrating scRNA-seq (*51*) (patients with AD) and spatial transcriptomics (APP/PS1 mice) (*52*). After grouping, annotation, and differential gene analysis, single-cell transcriptomics analysis revealed significant down-regulation of BMAL1, EP300, and SP1 in AD-associated astrocytes, neurons, and oligodendrocytes (Fig. 7, A and B, and fig. S5, A to E). Spatial transcriptomic profiling delineated marked down-regulation of BMAL1 in cerebral nuclei, cortical subplate, hippocampal formation, isocortex, olfactory areas, and thalamus of APP/PS1 mice (Fig. 7C and fig. S6A), while EP300 and SP1 exhibited down-regulation specifically within hippocampal formation, isocortex, and thalamus (fig. S6, B and C). Accumulating studies reported that the deficiency of BMAL1, an irreplaceable clock gene that governs multiple important physiological processes, promotes AD (*53*). Our data showed that the mRNA level of *Bmal1* was significantly up-regulated by MLT treatment in both APPswe cells and APP/PS1 mice (Fig. 7, D and E). MLT intervention significantly up-regulated BMAL1 expression compared to age-matched APP/PS1 controls (Fig. 7F). Immunofluorescence assays further demonstrated enhanced BMAL1 activation induced by MLT treatment particularly in the hippocampus and cortex (Fig. 7G). In addition, MLT elicited semblable BMAL1 activation in aged MSCs, evidenced by coordinated transcriptomic and translational up-regulation (Fig. 7, H to J). Further investigation confirmed that MLT activated the BMAL1 pathway in aging and AD pathology, accompanying the ascending of circadian locomotor output cycles kaput (CLOCK), leading to the activation of the transcription of core clock genes, including periods (PERs) and cryptochromes (CRYs), both in APPswe cells, APP/PS1 mice, and aged MSCs (Fig. 7, K to M). These results suggest that the positive effects of MLT may be attributed to the regulation of BMAL1-CLOCK signaling axis.

Subsequent genetic ablation of BMAL1 via small interfering RNA (siRNA) transfection abolished MLT-mediated suppression of SA-β-Gal activity in aged MSCs (Fig. 7N). BMAL1 knockdown abrogated MLT-induced inhibition of SASP expression, rescinded MLT-enhanced Lamin B1 transcriptional activation, and concomitantly elevated DNA damage marker H2AX phosphorylation in MLT-treated MSCs (Fig. 7, O to Q). In summary, our findings demonstrate that BMAL1 functions as a pivotal transcriptional effector mediating MLT's neuroprotective and antisenescence actions. By reactivating BMAL1-centered circadian gene expression through super-enhancer remodeling, MLT reprograms aging- and AD-related transcriptomic networks, offering a mechanistically grounded therapeutic strategy targeting epigenetic and temporal dysfunction in neurodegeneration.

DISCUSSION

It remains a critical unresolved question whether aging drives the onset of AD or whether AD pathology, in turn, accelerates biological aging. This fundamental question represents a critical challenge in the field and underscores the urgent need for therapeutics that can effectively target both aging and AD. Moreover, these therapeutics must be mechanistically robust, safe for long-term administration, and capable of modulating the multifactorial complexity of these intertwined pathologies. In this study, we addressed this unmet need by integrating an AI-driven PTD-DEP method with experimental validation, rapidly identifying MLT, an endogenous hormone used in insomnia treatment, as a promising dual-action candidate for AD and aging. Supported by clinical data from 161 human serum samples demonstrating a significant age-associated decline in MLT levels, behavioral and biochemical assessments confirmed that MLT restores cognitive function, mitigates neuroinflammation, and enhances synaptic plasticity in AD mouse models, consistent with previous reports. Notably, MLT also exhibited potent antisenescence effects in aged MSCs, including reduction of SA-β-Gal activity, inhibition of SASP, and alleviation of oxidative stress. Through PROTAC-based chemoproteomic analysis combined with CB-Dock2 prediction, p300 was identified as a principal molecular target of MLT. Subsequently, we used CUT&Tag, MAPE-PPI prediction, scRNA-seq, spatial transcriptomics, and advanced pharmacological techniques to explore and reveal that MLT alleviates aging and AD-related pathology by modulating BMAL1 transcription through p300/SP1-mediated super-enhancers.

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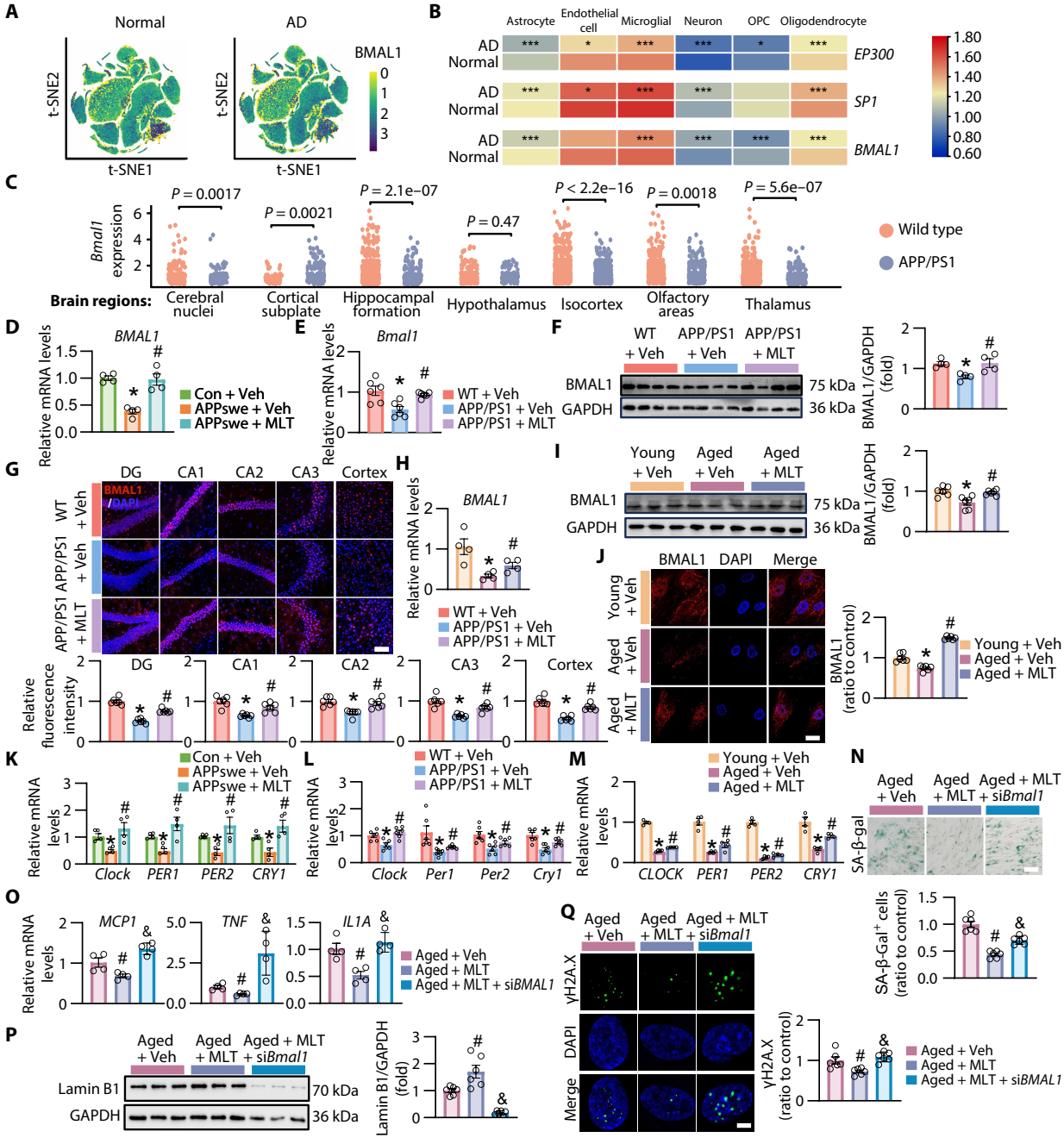


Fig. 7. BMAL1 functions as a central effector of MLT-mediated neuroprotection and antisenescence activity. (**A**) Scatter plot of *BMAL1* expression across cell types in patients with AD versus normal samples. (**B**) Heatmap showing significant changes in *EP300*, *SP1*, and *BMAL1* in patients with AD (**P* < 0.05; ****P* < 0.001). OPC, oligodendrocyte precursor cell. (**C**) Box plots depicting the down-regulation of *Bmal1* across multiple brain regions in APP/PS1 mice. (**D** and **E**) Relative mRNA levels of *BMAL1* in APPswe cells (D) and APP/PS1 mice (E) (*n* = 4 in APPswe cells and 6 in APP/PS1 mice group). (**F**) Western blot assays and quantification analysis of BMAL1 in APP/PS1 mice (*n* = 4 per group). (**G**) Representative immunofluorescent images and quantification analysis of BMAL1 in APP/PS1 mice (*n* = 6 per group). Scale bar, 50 μm. DG, Dentate gyrus. (**H**) Relative mRNA levels of *BMAL1* in MSCs (*n* = 4 per group). (**I**) Western blot assays and quantification analysis of BMAL1 in MSCs (*n* = 6 per group). (**J**) Representative immunofluorescent images and quantification analysis of BMAL1 in MSCs (*n* = 6 per group). Scale bar, 50 μm. (**K** to **M**) Relative mRNA levels of core clock genes (*CLOCK*, *PER1*, *PER2*, and *CRY1*) in APPswe cells (K), APP/PS1 mice (L), and MSCs (M) (*n* = 4 in APPswe cells and MSCs and 6 in APP/PS1 mice group). (**N**) Representative immunofluorescent images and quantification analysis of SA-β-Gal in MSCs (*n* = 6 per group). Scale bar, 100 μm. (**O**) Relative mRNA levels of SASPs (*MCP1*, *TNF*, and *IL1A*) (*n* = 4 per group). (**P**) Western blot assays and quantification analysis Lamin B1 in MSCs (*n* = 6 per group). (**Q**) Representative immunofluorescent images and quantification analysis of γH2AX foci in MSCs (*n* = 6 per group). Scale bar, 10 μm. Data are shown as means ± SEM. **P* < 0.05 versus control, WT, or young group; #*P* < 0.05 versus APPswe, APP/PS1, or aged group; &*P* < 0.05 versus MLT treatment group.

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AD pathogenesis involves multifactorial mechanisms, with aging recognized as the predominant risk factor. The extensive molecular and cellular pathways shared between aging and AD suggest that targeting aging processes offers a strategic avenue to mitigate AD pathology. However, drug development for aging and AD remains hindered by the intricate biological complexities, resulting in prolonged timelines, high costs, and low success rates. In recent years, AI techniques have garnered increasing attention for their capacity to integrate complex pathological networks, accelerate drug discovery, and reduce associated expenditures. Here, we present our AI-driven PTD-DEP model, trained on large-scale molecular structures and transcriptomic datasets, which enables quantitative prediction of drug candidates' multitarget profiles encompassing both AD therapeutic efficacy and antisenescence potential. This methodology profoundly streamlines multitarget pharmacotherapy identification, with MLT as an illustrative example. To further assess the robustness of PTD-DEP, we conducted randomized validation of four candidate drugs, including two high-scoring and two low-scoring (data S1), to benchmark our AI model accuracy and establish MLT's therapeutic primacy (fig. S7, A to E). Experimental results confirmed high model accuracy with persistent discordant predictions. For instance, eplivanserin was misclassified as lacking antiaging activity, yet experimental data confirmed partial efficacy in suppressing SASPs. Subsequent model optimization will expand training databases and refine parameters. Critically, experimental validation confirms MLT as the foremost dual anti-AD and antiaging candidate, underscoring its therapeutic priority. This further validates our two-step framework's capacity to minimize misclassification. Our findings demonstrate that the convergence of AI and pharmacology facilitates a paradigm shift in complex disease drug discovery, enabling rapid identification and rigorous validation of effective therapeutics.

Clinical studies demonstrate that patients with AD exhibit progressive MLT deficiency compared to age-matched controls, with cerebrospinal fluid MLT depletion correlating with pathological staging of disease severity (*54*). Our clinical data corroborate a progressive, age-related decline in serum MLT, underscoring its critical role in neuroprotection during aging. Preclinical investigations have revealed MLT-mediated cognitive improvement (*55*), Aβ deposition reduction (*56*), synaptic plasticity restoration (*57*), and anxiety/depression-like behavior amelioration (*58*). Consistently, our APP/PS1 transgenic mouse model confirms that MLT mitigates AD pathology by rescuing cognitive deficits and attenuating multifactorial neurodegenerative hallmarks, including Aβ accumulation, neuroinflammation, and oxidative damage. To explore MLT's antiaging mechanisms, we used MSCs as a senescence model, where MLT effectively suppressed aging hallmarks such as SA-β-Gal activity, SASP secretion, p16/p21/p53 signaling, oxidative stress, and mitochondrial dysfunction. Despite encouraging preclinical evidence, notable AD heterogeneity, together with variability in dosage and formulation, contributes to the contradictory clinical outcomes of MLT, underscoring the need for precision pharmacotherapy to guide its clinical translation (*59*–*62*). Collectively, these findings position MLT as a safe and efficacious candidate for targeting both AD and aging-related pathologies. Its therapeutic potential for cognitive impairment in elderly patients is promising, either as monotherapy or in combination regimens. Nevertheless, the precise molecular mechanisms underlying MLT's neuroprotective effects require further elucidation.

PROTAC technology, by inducing selective protein degradation, provides a powerful tool for precise target deconvolution (*42*). Here, we synthesized an M-PROTAC and used it in integrated proteomic analyses to elucidate MLT's molecular targets. Coupled with CB-Dock2 computational docking, we conclusively identified p300 as the principal target mediating MLT's therapeutic actions in AD and aging. This validates PROTAC-based target identification as a cutting-edge platform for mechanistic insight. The EP300 catalyzes H3K27ac, a hallmark of active SEs that drive target gene transcription (*63*). Our integrated IP-MS and co-IP experiments confirmed the physical interaction between p300 and SP1, forming a transcriptional regulatory complex orchestrating downstream gene expression. Further, MAPE-PPI prediction combined with multiomics profiling via CUT&Tag and ChIP-qPCR demonstrated co-occupancy of p300 and SP1 at the BMAL1 super-enhancer locus, with MLT dynamically modulating this complex to activate BMAL1 transcription. Collectively, these findings elucidate the molecular framework underpinning MLT's dual anti-AD and antiaging efficacy and establish a robust preclinical foundation for its translational development in neurodegenerative disease therapeutics.

Sleep disturbances disrupt neural-immune-endocrine homeostasis, promoting the development of multisystem pathologies spanning psychiatric comorbidities and chronic degenerative conditions including neurodegenerative disorders, cardiovascular diseases, and diabetes mellitus (*64*). The age-dependent prevalence escalation of sleep disorders highlights their pathophysiological relevance in aging populations. Accumulating evidence suggests that sleep disturbances may contribute to cognitive decline, promoting the development of AD pathology (*31*). Sleep disruption aggravates amyloid deposition, neuroinflammation, oxidative stress, and mitochondrial dysfunction, while aging and neurodegeneration, in turn, worsen sleep disturbances, together forming a self-reinforcing vicious cycle (*22*). The previous Mendelian randomization analysis leveraging publicly available genome-wide association study (GWAS) further established a causal relationship between insomnia and AD, suggesting that pharmacological modulation of sleep disorders may offer novel therapeutic avenues for AD therapies. Given our revealed causality between insomnia and AD, we prioritized dual anti-AD and antiaging agents among extant anti-insomnia drugs to accelerate precision therapeutics for patients with AD with comorbid sleep disorders. Our findings underscore the translational significance of MLT, a clinically approved agent for sleep modulation, as an effective modulator of both aging-associated and AD-related phenotypes. By elucidating the molecular circuitry linking circadian rhythm disruption to neurodegeneration, our study supports the development of circadian-optimized therapeutic strategies and positions MLT as a viable candidate for the integrated management of age-related neurodegenerative diseases.

BMAL1 is a master regulator of circadian rhythms and sleep-wake homeostasis (*65*), and its dysregulation has emerged as a hallmark of AD pathology. Extensive experimental evidence indicates that loss of *Bmal1* function exacerbates AD progression by promoting Aβ deposition, tau hyperphosphorylation, oxidative stress, and neuroinflammation (*53*). Converging clinical and preclinical evidence shows that circadian rhythm disruption is common and clinically relevant in both aging and AD (*66*). In line with this link, MLT has been reported to ameliorate central nervous system phenotypes by restoring circadian timing (*67*, *68*). Our pharmacological and mechanistic investigations reveal that MLT significantly up-regulates BMAL1 and core circadian regulators, including CLOCK, across both cellular and animal models of aging and AD. Further, MLT activates BMAL1 transcription via p300/SP1-driven super-enhancer assembly, thereby

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restoring the circadian transcriptional feedback loop. Collectively, these findings demonstrate that MLT exerts multimodal therapeutic effects in AD and aging by targeting the BMAL1 axis, supporting its potential as a circadian-modulating adjunctive therapy for neurodegenerative disorders.

In conclusion, we first leveraged an AI-driven method to rapidly and efficiently identify MLT as a dual-action candidate with both antiaging and anti-AD potential, notably reducing the time, labor, and costs typically associated with drug repurposing. By integrating interdisciplinary approaches, including PROTAC-based chemoproteomics, multiomics, and advanced computational analyses, we precisely delineated p300 as the principal molecular target and unraveled the mechanistic pathway by which MLT exerts its therapeutic effects via the p300/SP1/BMAL1 regulatory axis. We validated these findings in human clinical samples from aging and patients with AD, underscoring the translational significance of MLT as a safe and effective intervention. Collectively, our study offers a robust and interdisciplinary paradigm for the rational discovery, mechanistic dissection, and clinical translation of therapeutics targeting the complexities of aging-related diseases. This cross-disciplinary workflow therefore constitutes a reproducible blueprint for decoding the biological complexity of aging and accelerating the development of multitarget therapeutics.

Several limitations of this study warrant consideration. First, while MLT exhibited robust neuroprotective and antiaging efficacy in preclinical models, its therapeutic potential remains to be validated in rigorous, large-scale clinical trials. Our insomnia-focused screening inherently constrains MLT's therapeutic benefit in patients with AD with intact circadian rhythmicity. In future clinical applications, MLT, either as monotherapy or in combination with other agents, may be particularly suitable for patients stratified by dysregulation of the p300/SP1/BMAL1 axis, thereby supporting mechanism-informed patient stratification for targeted therapeutic application. Comprehensive evaluation of MLT as both an adjunctive and combination therapy is required to establish its clinical efficacy and safety in AD and aging populations. Second, although our AI-driven PTD-DEP framework demonstrated high predictive accuracy across diverse preclinical datasets, its generalizability and methodological robustness must be further assessed in broader patient cohorts and under varying pathophysiological conditions. The multifactorial interactome of AD pathogenesis intrinsically constrains the training therapeutic drug coverage. Future research should prioritize the translational validation of MLT in well-designed clinical settings, while simultaneously pursuing systematic extension and methodological refinement of the AI platform to accelerate the discovery of multitarget therapeutics for other complex neurological disorders. Further iterations of PTD-DEP should incorporate patient heterogeneity, including age, circadian profiles, and disease stage, to enable dose- and pathophysiological context–specific prediction of drug efficacy. We are upgrading the next PTD-DEP release to improve explainability and cross-disease applicability.

MATERIALS AND METHODS

General chemistry methods

All reactions were performed using a Teflon-coated magnetic stir bar at the indicated temperature and were conducted under an inert atmosphere when stated. All chemicals were used as received. Reactions were monitored by thin-layer chromatography (TLC) (silica gel 60 with fluorescence F254, visualized with a short-wave or long-wave ultraviolet lamp). Flash chromatography was conducted on silica gel using an automated system with detection wavelengths of 254 and 280 nm. All reported yields are isolated yields. All compounds are >95% pure by high-performance LC (HPLC). ^{1}H nuclear magnetic resonance (NMR) spectra were recorded on a Bruker AVANCE 300 MHz, AVANCE NEO 400 MHz, and AVANCE III HD 600 MHz. Chemical shifts (δ) are reported as parts per million relative to the residual undeuterated solvent as an internal reference. The abbreviations s, br s, d, t, q, dd, dt, ddd, and m stand for singlet, broad singlet, doublet, triplet, quartet, doublet of doublets, doublet of triplets, doublet of doublet of doublets, and multiplet, respectively. The purity of the target derivative M-PROTAC (>95%) was determined on a Nexera LC-40-PDA; Ultimate UHPLC AQ-C18, 1.8 μm, 3.0 mm by 150 mm; mobile phase, water/acetonitrile; flow rate, 0.5 ml/min; column temperature, 35°C; injection volume, 10 μl.

General procedure to synthesize M-PROTAC

Synthesis of N-(2-(5-hydroxy-1H-indol-3-yl)ethyl)acetamide

MLT (2.00 g, 8.61 mmol) was dissolved in anhydrous dichloromethane (DCM; 20 ml) under argon atmosphere. BBr_3 (1 M in DCM, 25.83 ml, 25.83 mmol) was dropped into the solution the reaction mixture at −40°C with stirring. After 1 hour, the reaction mixture was transferred to room temperature and stirred for 6 hours. After the reaction completion as monitored by the TLC, cold methanol was slowly added to the solution at −20°C to quench the reaction. Then, the solvent was removed under reduced pressure and neutralized with $NaHCO_3$ to pH 7. The aqueous phase was extracted with ethyl acetate (EtOAc) (×3), and the combined organic extracts were dried over sodium sulfate. The residue was concentrated to dryness in vacuo and purified by column chromatography on silica gel to afford *N*-(2-(5-hydroxy-1*H*-indol-3-yl)ethyl)acetamide (M1) as a brown syrup (1.43 g, 76.10% yield). ^{1}H NMR [300 MHz, dimethyl sulfoxide (DMSO)–d_6] δ 10.49 (s, 1*H*), 8.60 (s, 1*H*), 8.02 to 7.88 (m, 1*H*), 7.13 (s, 1*H*), 7.03 (d, J = 2.5 Hz, 1*H*), 6.81 (d, J = 2.4 Hz, 1*H*), 6.58 (dd, J = 8.6 and 2.3 Hz, 1*H*), 3.33 to 3.20 (m, 2*H*), 2.70 (t, J = 7.6 Hz, 2*H*), and 1.80 (s, 3*H*) (fig. S8).

Synthesis of 2-((3-(2-acetamidoethyl)-1H-indol-5-yl)oxy) acetic acid

A solution of M1 (1.43 g, 6.55 mmol) in anhydrous DMSO (20 ml) was added potassium *tert*-butoxide (808.7 mg, 7.21 mmol), and the reaction was stirred under argon atmosphere. After a few minutes, methyl bromoacetate (0.683 ml, 7.21 mmol) was added to the solution, and the reaction mixture was stirred at room temperature for 2 hours. After reaction completion as monitored by the TLC, the solution was diluted with EtOAc and H_2O, and the aqueous phase was extracted with EtOAc (×3). The combined organic extracts were washed with brine, dried over sodium sulfate, and concentrated to a brown oil. The brown oil was dissolved in a solution of tetrahydrofuran (THF) and H_2O (20 ml; THF:H_2O = 1:1) at room temperature. LiOH (212.6 mg, 8.8 mmol) was added in the solution at 0°C and stirred for 1 hour. The reaction solution was diluted with H_2O (20 ml) and washed with DCM (×3). The aqueous phase was neutralized to pH 7 with 6 M HCl and was extracted with EtOAc (×3). Then, it was concentrated to 10 ml in vacuo and placed in a refrigerator at 4°C overnight. The next day, we filtered the solution and purified it by flash column chromatography to afford 2-((3-(2-acetamidoethyl)-1*H*-indol-5-yl)oxy)acetic acid (M2) (1.20 g, 66.29% yield) as the brown solid. ^{1}H NMR (400 MHz, DMSO-d_6) δ 12.85 (s, 1*H*), 10.67

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(d, J = 2.4 Hz, 1*H*), 7.92 (t, J = 5.8 Hz, 1*H*), 7.23 (d, J = 8.7 Hz, 1*H*), 7.11 (d, J = 2.6 Hz, 1*H*), 6.98 (d, J = 2.5 Hz, 1*H*), 6.74 (dd, J = 8.7 and 2.5 Hz, 1*H*), 4.62 (s, 2*H*), 3.29 to 3.25 (m, 2*H*), 2.75 (t, J = 7.3 Hz, 2*H*), and 1.80 (s, 3*H*) (fig. S9).

Synthesis of tert-butyl(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)amino)ethoxy)ethyl)carbamate

N-Boc-2-(2-aminoethoxy)ethanamine (2.17 ml, 10.86 mmol) was added to the solution of 2-(2,6-dioxopiperidin-3-yl)-4-fluoroisoindoline-1,3-dione (2.00 g, 7.24 mmol) in *N*,*N*′-dimethylformamide (DMF). Followed by *N*,*N*-diisopropylethylamine (DIPEA; 3.74 g, 8.96 mmol), the mixture was stirred at 90°C overnight. After reaction completion as monitored by the TLC, the reaction mixture was diluted with H_2O and extracted with EtOAc (×3). The organic extracts were washed with brine and dried over sodium sulfate. The solvent was removed in vacuo and purified by column chromatography to afford *tert*-butyl(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)amino)ethoxy)ethyl)carbamate (N1) as yellow-green syrup (2.13 g, 63.88% yield). ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.13 (s, 1*H*), 7.50 (dd, J = 8.5 and 7.1 Hz, 1*H*), 7.11 (d, J = 7.1 Hz, 1*H*), 6.92 (d, J = 8.5 Hz, 1*H*), 6.51 (s, 1*H*), 4.97 (s, 1*H*), 4.92 (dd, J = 12.5 and 5.4 Hz, 1*H*), 3.68 (t, J = 5.3 Hz, 2*H*), 3.55 (t, J = 5.2 Hz, 2*H*), 3.46 (t, J = 5.4 Hz, 2*H*), 3.33 (d, J = 5.8 Hz, 2*H*), 2.91 to 2.67 (m, 3*H*), 2.12 (dtd, J = 12.5, 4.9, and 2.4 Hz, 1*H*), and 1.42 (s, 9*H*) (fig. S10).

Synthesis of 4-((2-(2-aminoethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

To the above solution in dioxane, hydrochloric acid (10 ml, 4 M in dioxane) was slowly added dropwise under an ice bath, followed by stirring at room temperature for 2 hours, and solid gradually precipitated out. Then, we filtered the solid to obtain hydrochloride of 4-((2-(2-aminoethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl) isoindoline-1,3-dione (N2) as a yellow solid (1.50 g, 89.99% yield) without extra purified.

Synthesis of 2-((3-(2-acetamidoethyl)-1H-indol-5-yl)oxy)-N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl) amino)ethoxy)ethyl)acetamide (M-PROTAC)

M2 (0.10 g, 361.94 μmol) was dissolved in DMF, followed by the addition of DIPEA (157.61 ml, 904.84 μmol) and hexafluorophosphate azabenzotriazole tetramethyl uronium (204.80 mg, 542.90 μmol) in sequence. The reaction was stirred at 0°C for 1 hour. Hydrochloride of N2 (143.60 mg, 361.94 μmol) was added into the reaction and stirred at room temperature for 12 hours. After reaction completion as monitored by the TLC, the solution was diluted with EtOAc and H_2O, and the aqueous phase was extracted with EtOAc (×3). The combined organic extracts were washed with brine and dried over sodium sulfate. The resulting solution was concentrated in vacuo and was purified by flash column chromatography to obtain M-PROTAC (0.096 g, 42.87% yield). ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.08 (s, 1*H*), 10.68 (d, J = 2.4 Hz, 1*H*), 8.03 (t, J = 5.8 Hz, 1*H*), 7.91 (t, J = 5.7 Hz, 1*H*), 7.57 (dd, J = 8.6 and 7.0 Hz, 1*H*), 7.23 (d, J = 8.8 Hz, 1*H*), 7.12 (d, J = 8.7 Hz, 2*H*), 7.05 (d, J = 2.4 Hz, 1*H*), 7.03 (d, J = 7.0 Hz, 1*H*), 6.79 (dd, J = 8.8 and 2.5 Hz, 1*H*), 6.60 (t, J = 5.7 Hz, 1*H*), 5.03 (dd, J = 12.9 and 5.5 Hz, 1*H*), 4.44 (s, 2*H*), 4.09 (s, 1*H*), 3.60 (t, J = 5.4 Hz, 2*H*), 3.52 (t, J = 6.0 Hz, 2*H*), 3.45 (t, J = 5.5 Hz, 2*H*), 3.36 (s, 2*H*), 3.31 to 3.23 (m, 2*H*), 3.17 (s, 2*H*), 2.90 to 2.78 (m, 1*H*), 2.79 to 2.72 (m, 2*H*), 2.60 to 2.52 (m, 1*H*), 2.46 (dd, J = 13.1 and 4.4 Hz, 1*H*), 2.04 to 1.93 (m, 1*H*), and 1.79 (s, 3*H*). ^{13}C NMR (151 MHz, DMSO-d_6) δ 172.70, 169.99, 168.96, 168.90, 168.30, 167.24, 151.28, 146.38, 136.18, 132.05, 131.81, 127.44, 123.52, 117.38, 111.90, 111.72, 111.34, 110.64, 109.25, 102.07, 68.70, 68.63, 68.09, 48.53, 41.63, 39.33, 38.07, 30.91, 25.14, 22.64, and 22.08. High-resolution MS (electrospray ionization) mass/charge ratio (m/z): calculated for $C_{31}H_{34}N_6O_8$ $[M + H]^+$ 619.25109, found 619.25258 (figs. S11 to S14).

Molecular docking

Small-molecule structures were retrieved from PubChem, and protein structures were retrieved from UniProt. Docking was performed on the free, publicly accessible CB-Dock2 server using default parameters, and results were visualized in PyMOL.

Biological pathway prediction module of PTD-DEP

Data collection

Using methods described in earlier studies (*69*), we extracted four pathway sets—anti-AD, antiaging, anti-inflammation, and antioxidation)—from the STRING database to construct a gene expression network. Key hub genes in each pathway were pinpointed with the CytoHubba plug-in for Cytoscape. We then mined PubChem BioAssays for compounds that not only target these hub genes but also show confirmed bioactivity. Specifically, we retrieved gene-centric bioactivity assay records from PubChem, kept only entries with explicit labels (active and inactive), and binarized outcomes at the pathway level (active = 1 and inactive =0). Assays without definitive labels were excluded. Following data curation, we assembled 17,598 definitively labeled bioactivity assay entries spanning four pathways. Because inadequate BBB penetration often limits therapeutic efficacy in AD, we additionally gathered small molecules with experimentally verified BBB permeability (*70*). This dataset covers BBB permeability data for 2035 compounds.

To build an external, prospective screening library used only for posttraining prioritization, we queried DrugBank with the keyword "insomnia," retrieving 67 compounds with available structures. We then searched DrugBank's anatomical therapeutic chemical section using "N05C (hypnotics and sedatives)" as the keyword and identified 64 compounds with publicly available structures. We merged the results from the two approaches. As keyword searches can surface agents reported to induce insomnia, each hit was manually checked against the primary DrugBank annotations and the peer-reviewed literature. Compounds associated with insomnia induction and any duplicates were removed, yielding a final set of 92 medicines. This library was excluded from all model training and evaluation to prevent data leakage.

Model training

To prevent leakage, any compound present in the screening library was removed from the model development pool. All datasets were subjected to compound-level splitting (80% training and 20% held-out test) with a fixed random seed, ensuring that no molecule appears in more than one split. Molecules were represented by extended-connectivity fingerprint (ECFP4; radius = 2, 2048-bit) fingerprints using RDKit and used as input features for model training. Six classifiers—Decision Tree, Random Forest, Logistic Regression, XGBoost, Gradient Boosting, and CatBoost—were tuned in a unified pipeline. For models with fewer hyperparameters or well-understood tuning behavior (Logistic Regression, DecisionTree, and GradientBoosting), we used GridSearchCV with fivefold cross-validation to exhaustively evaluate all parameter combinations. For models with larger or continuous parameter spaces (CatBoost, XGBoost, and RandomForest), we used RandomizedSearchCV with 30 iterations to balance performance and computational cost. Model performance was assessed using the ROC-AUC. The hyperparameter set with the

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highest average ROC-AUC was selected as optimal, and the corresponding model was retained as the final model (table S1).

Drug screening strategy

Using the PTD-DEP machine learning layer, we generated 0/1 labels for five properties in 92 anti-insomnia agents: anti-AD, antiaging, anti-inflammation, antioxidation activity, and BBB permeability. We then selected the nine highest-scoring BBB-permeable compounds for subsequent screening.

Transcriptomics prediction component of PTD-DEP

Data collection

We compiled a training set comprising 1,319,138 L1000 expression profiles generated by 43,526 small molecules in 76 cell lines. To ensure brain-relevant predictions, we drew on the Human Cell Atlas scRNA-seq resource, which contained 747,366 cells (*30*). After batch correction, dimensionality reduction, and reannotation, we obtained 61 well-defined cell types. Residual cross-platform batch effects between L1000 and single-cell data were removed following published procedures (*71*). We then calculated cell type and cell line centroids and built similarity maps with cosine similarity, Euclidean distance, and Pearson correlation. All three metrics converged on four cell lines—HCT116, HEK293T, JURKAT, and NPC—as the closest transcriptomic surrogates for human brain cells; their perturbation profiles were retained for model development. The dataset was randomly split 80:20 into development set and held-out test set. The development set was further split 80:20 into training and validation set (64%:16%:20% overall). Preprocessing was fit on the training set; model selection used validation R^2; the held-out test set was evaluated exactly once.

Feature engineering

Molecules were represented by ECFP4 (radius = 2, 2048-bit) fingerprints, while cell line identity was represented by one-hot vectors.

Linear regression baseline

Building on the dataset partitioning and feature extraction approach described above, a baseline linear regression model was used to predict compound molecular perturbations, achieving averaged sample-wise $R^2 = 0.2563$ on the held-out test set.

Model architecture

An autoencoder was trained to compress high-dimensional expression data into a compact latent space. Compounds were converted to molecular graphs with MolGraphConvFeaturizer from DeepChem (v2.8.0). Graph-based features and cell line embeddings were concatenated and fed into a graph convolutional network, followed by fully connected layers to predict transcriptomic responses.

Performance

After training, the model reproduces compound-induced expression profiles with an averaged sample-wise $R^2 = 0.7799$ on the held-out test set. To approximate biological replicates and improve robustness, each perturbation was predicted eight times. All analyses were carried out in Python 3.11.9.

Mendelian randomization analysis

We retrieved insomnia GWAS summary statistics (ebi a GCST-90018649) and AD GWAS data (finn b F5_ALZHDEMENT) from the integrative epidemiology unit (IEU) OpenGWAS repository. Using insomnia as the exposure, we retained variants with $P < 1 \times 10^{-4}$ and pruned them for linkage disequilibrium in TwoSampleMR v0.4.2.6 ($r^2 < 1 \times 10^{-4}$, 100,000-kb window). Variants with an F statistics of <10 were excluded to maintain instrument strength. Twenty-nine single-nucleotide polymorphisms passed all filters and were included in the Mendelian randomization analysis. All procedures were run in R 4.4.3.

Measurement of MLT in serum

All participants provided written informed consent, and the study protocol, including recruitment strategies and ethical considerations, were approved by the Ethics Committee of National Cancer Center/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College with the number 21/197-2868. Detailed information of all participants is presented in table S2.

An equal volume of plasma from each sample was mixed with five volumes of prechilled methanol to precipitate proteins. The mixture was vortexed thoroughly and incubated at 4°C for 30 min, followed by centrifugation at 20,000*g* for 15 min. The supernatant was carefully transferred into a new centrifuge tube and then concentrated and evaporated. A small amount of methanol is used to redissolve the sample and then transferred to the sample vials for subsequent analysis.

The concentrations were assessed using ultrahigh-performance LC (UHPLC) with mass spectrometer detection. Chromatographic separation was performed using UHPLC LC-30A (Shimadzu, Tokyo, Japan). An injection volume of 1 μl of processed samples was separated on ACQUITY UPLC HSS T3 (1.8 μm, 2.1 mm by 100 mm; sourced from Waters) maintained at 40°C. The total flow rate was 0.3 ml/min. Solvent A consisted of LC-MS grade water containing 0.1% formic acid, and solvent B consisted of LC-MS grade acetonitrile. The gradient elution program is as follows: 0 to 1 min, 5% B; 1 to 6 min, increase to 95%; 6 to 8 min, 95%; 8 to 8.1 min, decrease to 5%; and, lastly, 8.1 to 10 min, 5% for reconditioning the column.

MLT was quantitatively analyzed using a SCIEX QTRAP 5500 mass spectrometer in MRM mode with positive ionization. The MS conditions were as follows: An ion spray voltage was set to 5500 V, a turbo spray temperature was set to 550°C, and nebulizer and heater gas flows were both set to 55 arbitrary units. The curtain gas was maintained at 35 arbitrary units, and the interface heater was activated. The declustering potential and collision energy were optimized to 120 V and 20 eV, respectively, for MLT. Nitrogen was used as the collision gas in all cases. The MS transitions for quantification were optimized as follows: MLT at m/z 233 → 174.

Cell culture and treatment

All cell lines used in this study were maintained according to standard protocols provided by the American Type Culture Collection. SH-SY5Y and N_2A cells were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin (100 U/ml)/streptomycin (100 μg/ml). MSCs were grown in DMEM/F12 containing 10% FBS and 1% penicillin/streptomycin. All cells were incubated under standard conditions (5% CO_2 atmosphere at 37°C) and passaged at a 1:3 ratio upon reaching 80 to 90% confluency.

Cells were seeded into culture flasks, and when the cell density reached about 50%, the addition of virus to construct cell lines was started. Negative control virus and APP695 (K595N and M596L) virus were transfected at the same concentration of 4 × 10 TU/ml. Puromycin (2 μg/ml) was added to the cells 72 hours after virus transfer, and the subsequent cell culture medium contained puromycin.

Therapy was performed by adding MLT to cell complete medium at a treatment concentration of 400 μM for 48 hours. For p300

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inhibition, APPswe cells or MSCs were pretreated with C646 (25 μM) for 4 hours. For BMAL1 gene silencing, cells were transfected with specific siRNA using the following protocol: Culture medium was replaced with serum-free Opti-MEM (Thermo Fisher Scientific) before transfection. The transfection complex was prepared by separately diluting 200 nM BMAL1-targeting siRNA (forward, 5′-CCGAGGGAAGAUACUCUUUTT-3′; reverse, 5′-AAAGAGUAUCUUCCCUCGGTT-3′) and Lipofectamine 3000 reagent (Thermo Fisher Scientific) in Opti-MEM, followed by gentle combination of the two solutions at a 1:1 volume ratio. The mixture was incubated at room temperature for 15 to 20 min to allow complex formation. Subsequently, cells were washed with phosphate-buffered saline (PBS) and overlaid with the transfection complex/Opti-MEM solution. Following 12-hour incubation under standard culture conditions (37°C under 5% CO_2), the transfection medium was replaced with complete growth medium supplemented with 10% FBS. Functional assays or immunostaining was performed 48 hours posttransfection to assess knockdown efficiency.

Animals and manipulations

All WT or APP/PS1 male C57/BL6J mice (aged 6 months) were maintained in a pathogen-free facility under a 12-hour light–12-hour dark cycle at 22° ± 1°C with free access to regular chow and water. All animal experiments were carried out in accordance with the principles provided by the National Institute of Health Guideline and were approved by the Animal Care and Use Committee of China Pharmaceutical University (ethics number: 2023-08-001). Mice were assigned to three experimental groups ($n = 10$ per group): WT (C57BL/6J), APP/PS1 transgenic model, and APP/PS1 + MLT treatment. Twenty male APP/PS1 transgenic mice were randomly assigned to two groups in a double-blind design. APP/PS1 mice in one group were treated with MLT dissolved in pure water containing 1% ethanol at a concentration of 100 μg/ml for 1 month, and the other mice were also given 1% ethanol in pure water as control.

RNA isolation and reverse transcription qPCR analysis

For mouse organ tissues and cells, total RNA was extracted with RNA isolater Total RNA Extraction Reagent (Vazyme). RNA was reverse transcribed using HiScript II Reverse Transcriptase (Vazyme) in GeneExplorer (Bioer). Real-time PCR was performed on CFX Opus 384 (Bio-Rad), cycled in 95°C for 15 s, 60°C for 30 s, and 72°C for 30 s for 35 cycles, after using SYBR Green PCR Master Mix (Vazyme) for an initial denaturation step at 95°C for 10 min. Relative quantitative method was used to compared the transcript quantities. The amount of detected mRNA was normalized to the amount of endogenous control. The relative value to the control sample is given by $2^{-\Delta\Delta CT}$. The specific primer sequences for reverse transcription PCR were listed in table S3.

Western blot

Protein lysates were obtained from N_2A cells, APPswe cells, MSCs, or brain tissues. The total protein concentration was determined by the bicinchoninic acid (BCA) Protein Assay Kit (Beyotime). Protein (15 to 30 μg) was separated by 8 to 12% SDS–polyacrylamide gel electrophoresis (PAGE) gel initially and then transferred to polyvinylidene difluoride (PVDF) membranes. Afterward, PVDF membranes were sealed with 5% nonfat powdered milk for 2 hours and incubated with primary antibodies (table S4) at 4°C overnight. The next day, membranes were incubated with horseradish peroxidase–conjugated anti-rabbit or mouse immunoglobulin G (IgG) (table S4) at room temperature for 2 hours. Last, after being washed three times, membranes were imaged by ChemiDoc System (Bio-Rad) to analyze the signal intensities.

Morris water maze

After treatment, the learning and memory function of mice in each group was detected by MWM. The MWM experimental system consists of a water maze and a behavioral recording and analysis system. The water maze includes a circular water pool with a radius of 1 m, a circular platform with a diameter of 10 cm, and a temperature control device. The water in the pool is dyed white with titanium dioxide (to highlight the C57BL/6J mice), and the water surface is divided into four quadrants. The platform is located in the middle of the third quadrant and is submerged 1 cm below the water surface. The temperature control system maintains the water temperature at approximately 25°C. In addition, different markers are distributed around the water maze to help the mice identify directions and the location of the platform. The MWM test is conducted over 7 days. Day 0 is the adaptation period, allowing the mice to adapt to the water maze environment. Days 1 to 5 are the learning period, during which the mice are placed in the water from the first quadrant and then allowed to freely explore for 90 s. The latency to reach the platform is recorded. For mice that fail to find the platform, they are guided to the platform for 30 s of learning, and the latency is recorded as the maximum value of 90 s. Day 6 is the formal test period. On the basis of the previous 6 days, the platform is removed, and the movement trajectory of the mice, as well as the swimming distance and time in the target quadrant, is recorded.

Thioflavin S staining

Thioflavin S staining was used to detect Aβ plaques and deposits in brain sections of APP/PS1 mice. The brain slices were washed three times with PBS and then incubated in the dark with 0.005% thioflavin S staining solution for 10 min. Subsequently, they were washed twice with 50% ethanol for 5 min in succession, followed by three consecutive washes with PBS for 5 min each. Last, they were incubated in the dark with 4′,6-diamidino-2-phenylindole (DAPI) for 10 min and were ready for mounting. The brain slices were transferred onto glass slides and sealed with an antifluorescence quencher. The samples were imaged using a fluorescence microscope (BioTek, Cytation5), and the obtained images were analyzed and quantified using ImageJ software.

ROS detection

For brain tissue samples, the Reactive Oxygen Species Assay Kit for Superoxide Anion with DHE (Beyotime) was used to measure the ROS level in APP/PS1 mouse brain slices. The brain slices were washed three times with PBS and then treated with 0.3% Triton X-100 for 20 min. Subsequently, they were stained with 10 μM DHE staining solution and incubated at 37°C in the dark for 30 min. After washing three times with PBS, they were incubated with DAPI in the dark for 10 min and were ready for patching. The brain slices were transferred to a glass slide, and antifluorescence quencher was added for sealing. The samples were imaged a fluorescence microscope (BioTek, Cytation5), and the obtained images were analyzed and quantified using ImageJ software.

For the cell samples, the Reactive Oxygen Species Assay Kit with CM-H2DCFDA (Beyotime) was used to detect the ROS level in senescent MSCs. One milliliter of 10 μM 2′,7′-dichlorodihydrofluorescein

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diacetate staining solution was added to each well of a six-well plate and incubated in a 37°C incubator for 20 min. Subsequently, the cells were washed three times with serum-free cell culture medium and were ready for photography. The samples were imaged using a fluorescence microscope, and the obtained images were analyzed and quantified using ImageJ software.

RNA-seq and analysis

Total RNA was extracted from the cortex of mice and MSCs, respectively, for RNA-seq analysis. RNA quality was assessed by Agilent Bioanalyzer 2100 system (Agilent Technologies). cDNA library construction and sequencing were performed by Novogene. Raw reads were trimmed with fastp (v 1.0.1) and aligned to the mouse (GRCm38) or human (GRCh38.p14) reference genome using HISAT2 (v2.2.1) (*72*, *73*). Gene-level counts were obtained with featureCounts (v 2.0.3), and differential expression was calculated in R (4.4.3) with limma (v3.5.1) or DESeq2 (v1.42.1) (*74*). All steps were executed on Ubuntu 18.04 LTS; exact command-line options are recorded in the accompanying code.

SA-β-Gal staining

Cellular SA-β-Gal was detected using the SA-β-Gal staining kit purchased from Beyotime. The culture medium of the treated cells was discarded, and the cells were washed twice with precooled PBS. Then, 1 ml of SA-β-Gal staining and fixation solution was added to each well of a six-well plate, and the cells were fixed at room temperature for 15 min. The SA-β-Gal staining and fixation solution was removed, and the cells were washed three times with 1 ml of PBS on a shaker at 60 rpm for 3 min each time. After removing the PBS, 1 ml of staining working solution (containing 10 μl of SA-β-Gal staining solution A, 10 μl of SA-β-Gal staining solution B, 930 μl of SA-β-Gal staining solution C, and 50 μl of X-galactosidase solution) was added to each well. The plate was sealed with parafilm and incubated overnight at 37°C. The samples were imaged using a fluorescence microscope (BioTek, Cytation5), and the obtained images were analyzed and quantified using ImageJ software.

JC-1 staining

After the cells were treated with the drug for 24 hours, the mitochondrial membrane potential was detected using the enhanced mitochondrial membrane potential detection kit (Beyotime). Following removal of the culture medium, adherent cells were gently washed two times with PBS (pH 7.4) to eliminate residual medium components. Cells were then incubated with JC-1 staining solution at 37°C under 5% CO_2 atmosphere for 20 min, protected from light. After incubation, the staining solution was carefully aspirated, and cells underwent two consecutive washes with JC-1 assay buffer to remove unbound dye. Last, fresh complete medium was added to maintain cell viability before immediate microscopic analysis. Mitochondrial membrane potential was assessed using a fluorescence microscope (BioTek, Cytation5) with dual-channel detection: 485-nm excitation/530-nm emission for monomeric JC-1 (green fluorescence) and 535-nm excitation/590-nm emission for J-aggregates (red fluorescence).

Immunofluorescence

Fresh brain tissues were isolated and then fixed in 4% paraformaldehyde dissolved in PBS for at least 24 hours. The brain tissues were dehydrated with 30% sucrose solution for 48 hours. Brain tissue slices with a thickness of 30 μm were cut by a freezing microtome (Leica, CM1950), and the slices with intact hippocampi were stored in the freezing solution (PBS:ethylene glycol:glycerin = 5:3:2) at −20°C.

Following three washes in PBS, brain slices were permeabilized with 0.3% Triton X-100 in PBS for 20 min at room temperature. Subsequently, nonspecific binding sites were blocked by incubating the slices with 5% bovine serum albumin (BSA) in PBS for 1 hour. After blocking, slices were incubated overnight at 4°C with target-specific primary antibodies (table S4) diluted in PBS containing 1% BSA. The following day, slices were washed three times with PBS (5 min per wash) and then incubated with fluorophore-conjugated secondary antibodies (table S4) for 1 hour under light-protected conditions. After additional PBS washes, nuclei were counterstained by incubating slices with DAPI (0.1 μg/ml; Beyotime, C1002) in PBS for 10 min in the dark. The brain slices were transferred onto glass slides, sealed with an antifluorescence quenching agent, and imaged using a laser confocal microscope (Zeiss, LSM800). The images obtained were analyzed and quantified using ImageJ software.

For cell samples, the cells were cultured in confocal culture dishes. After discarding the culture medium, they were washed three times with PBS. Subsequently, 4% paraformaldehyde was added to each well and fixed for 30 min. Similarly, it was washed three times with PBS. After treatment with 0.3% Triton X-100 for 20 min, it was blocked with 5% BSA for 1 hour. After the blocking is completed, it was incubated overnight at 4°C with the primary antibody that specifically binds to the target protein. The next day, after washing three times with PBS, it was incubated at room temperature in the dark for 1 hour with the corresponding fluorescent secondary antibody. Last, it was washed again with PBS, incubated in the dark for 10 min with DAPI, then added the antifluorescence quencher to the confocal dish, and ready for shooting. The samples were imaged using a laser confocal microscope (Zeiss, LSM800), and the obtained images were analyzed and quantified using ImageJ software.

Targeted degradation omics

N_2A cells were seeded into T25 culture flask, and when the cell density reached 70%, 10 μM MLT and 10 μM M-PROTAC were added, respectively, and placed in an incubator with constant temperature for 24 hours. After proteins extraction, BCA method was used to determine the protein concentration, and the subsequent protein samples were analyzed by LC-MS.

Cellular thermal shift assay

N_2A cells were seeded into 10-cm culture dishes and randomized into vehicle-treated and MLT-treated groups. At 70% confluency, cells were treated with 400 μM MLT (or an equivalent volume of DMSO vehicle) for 12 hours. Cells were then harvested using 0.25% trypsin-EDTA, pelleted by centrifugation (300*g* for 5 min), and washed three times with ice-cold PBS supplemented with 1 μM phenylmethylsulfonyl fluoride (PMSF). The cell pellet was resuspended in 880 μl of PMSF-containing PBS and evenly aliquoted into 11 samples.

Thermal denaturation was conducted using a thermal cycler with the following protocol: Samples were heated to designated temperatures (40°, 42°, 43°, 44°, 45°, 46°, 47°, 49°, 51°, 53°, and 55°C) for 3 min, followed by equilibration at 25°C for 3 min. Treated cells underwent 3 cycles of flash freezing in liquid nitrogen and thawing at 37°C. Lysates were clarified by centrifugation (12,000*g* for 15 min at 4°C). Supernatants were collected, mixed with 20 μl of 5× SDS loading buffer, denatured at 95°C for 10 min, and stored at −80°C before Western blot analysis.

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Cleavage Under Targets and Tagmentation

Nuclear extracts for CUT&Tag assays were obtained from mouse brain tissues using a nuclear extraction kit (Vazyme). The extracted nuclei were subjected to CUT&Tag experiments using the Hyperactive Universal CUT&Tag kit (Vazyme). First, at room temperature, the nuclei in the suspension were bound to magnetic beads coated with concanavalin A. Then, the cell membranes were permeabilized with 5% digitonin to allow the antibodies against p300, SP1, and H3K27ac (table S4) to fully bind to their target proteins inside the cells. After overnight incubation with the primary antibodies at room temperature, the secondary antibodies were added to the system and incubated for another 30 min. Subsequently, the DNA fragments bound to the target proteins were cleaved by high-activity pA-Tn5 transposons. The DNA fragments were then enriched and purified. The TruePrep Index Kit V2, based on the Illumina sequencing platform, was used to ligate the DNA fragments with P5 and P7 adapters, followed by PCR amplification to generate the DNA library.

After sequencing, we first used fastp to perform quality control on the CUT&Tag sequencing data. Then, the quality-controlled data were aligned to the GRCh38.p14 reference genome using bowtie2 (v 2.3.5.1) (*75*). After alignment, we used MACS2 (v 2.2.7.1) for peak calling to obtain the position information of the binding fragments. ChIPseeker was used to annotate the peaks. deepTools (v 3.5.6) and Integrative Genomics Viewer (IGV) (v 2.19.1) were used for visualization analysis (*76*). The specific parameters are detailed in the code, this part of the analysis was conducted in an Ubuntu 18.04-LTS environment, and full parameters are provided in the code repository.

Coimmunoprecipitation

Cortical tissue samples from mice were processed under nonreducing conditions. Briefly, tissues were gently lysed in ice-cold radioimmunoprecipitation assay lysis buffer (containing protease/phosphatase inhibitors) to extract total proteins. For IP, protein lysates were incubated overnight at 4°C with 2 μg of primary antibodies targeting p300 and SP1, along with species-matched nonspecific IgG (table S4) as an isotype control. The following day, 20 μl of prewashed Protein A/G magnetic beads (Thermo Fisher Scientific, 88802) were added to each sample and incubated for 1 hour at room temperature with gentle rotation to facilitate antibody-antigen complex binding.

Magnetic separation was performed using a magnetic rack (10 s), followed by careful removal of the supernatant. Beads were washed three times with 100 μl of inhibitor-supplemented lysis buffer (5 min per wash with agitation). After the final wash, bound protein complexes were eluted by resuspending beads in 25 μl of 5× SDS-PAGE loading buffer and denatured at 100°C for 15 min. Samples were briefly separated on a magnetic rack (10 s), and supernatants containing eluted proteins were collected for subsequent Western blot analysis. Alternatively, the magnetic beads were eluted with an acidic washing solution for subsequent IP-MS analysis.

IP-MS analysis

Protein digestion was performed using the filter-aided sample preparation method (*77*). Briefly, protein samples were dissolved in 8 M urea buffer and loaded into 10-kDa molecular weight cutoff ultrafiltration devices (Millipore). Reduction was achieved by adding 1 M dithiothreitol to a final concentration of 10 mM, followed by incubation at 35°C for 1.5 hours. Alkylation was subsequently performed in the dark by adding 0.5 M iodoacetamide to a final concentration of 20 mM, with 40-min incubation at room temperature. The buffer was exchanged with 50 mM ammonium bicarbonate (NH_4HCO_3; pH 8.0 to 8.5) through three successive centrifugation washes (14,000*g* for 20 min). Trypsin (sequencing grade, 2 μg) was added at a 1:50 (w/w) enzyme-to-protein ratio and digestion proceeded for 16 hours at 37°C. Peptides were collected by centrifugation and acidified with 0.1% formic acid before desalting using C18 StageTips. The eluted peptides were vacuum dried and reconstituted in 0.1% formic acid to a final concentration of 0.5 μg/μl.

For LC-MS/MS analysis, 2 μl of aliquots (1 μg of total) were loaded onto a nanoflow HPLC system (Thermo Fisher Scientific) coupled to an Orbitrap Exploris 480 mass spectrometer. Separation was achieved using an Acclaim PepMap RSLC C18 column (75 μm by 15 cm, 3 μm, 100 Å; Thermo Fisher Scientific) with a 90-min linear gradient from 3 to 35% acetonitrile in 0.1% formic acid at 300 nl/min. The mass spectrometer operated in data-dependent acquisition mode with the following parameters: MS1 scans at 60,000 resolution (m/z 200) over 350 to 1500 m/z range, followed by high energy collision dissociation fragmentation of the 20 most intense precursors using 30% normalized collision energy. MS2 spectra were acquired at 15,000 resolutions with dynamic exclusion set to 20 s. Protein identification was processed using Proteome Discoverer software (version 2.4, Thermo Fisher Scientific, Waltham, MA, USA) according to the protein database on the UniProt website.

For targeted degradomics, MaxQuant (version 1.6.17.0) was used to analyze the RAW files against the UniProt protein database. Search parameters included tryptic specificity with up to two missed cleavages, carbamidomethylation of cysteine as fixed modification, and methionine oxidation as variable modification. The false discovery rate was set to 1% at both peptide and protein levels. Label-free quantification intensities were $\log_2$ transformed and analyzed using Perseus software (version 1.6.15.0). Contaminant proteins and reverse hits were removed before statistical analysis.

DeepD2V

To build the model capable of accurately predicting downstream cobinding sites of p300 and SP1, we obtained p300 and SP1 ChIP-seq peaks from ReMap 2022 (BED format) (*78*). We adopted the published DeepD2V architecture. Negative sequences at a 3:1 ratio to positives were generated with genNullSeqs (gkmSVM v0.83.0; nMaxTrials = 60; xfold = 3) using the masked hg38 reference (BSgenome.Hsapiens.UCSC.hg38.masked). The resulting positive and negative sets for p300 and SP1 were used to train separate DeepD2V models. We set the learning rate to 5×10^{-4} and the batch size to 32, keeping all other hyperparameters at their defaults. In addition, the GRCh38 genome was partitioned into nonoverlapping 200-bp windows, of which 1,098,591 were randomly held out as a test set for predicting p300-SP1 cobinding regions.

Chromatin immunoprecipitation–quantitative polymerase chain reaction

The ChIP-qPCR experiment was used to detect the binding of p300, SP1, and H3K27ac modifications to the SE region of the BMAL1 gene. After culturing senescent MSCs to 80%, they were crosslinked and fixed at room temperature with 1% formaldehyde for 10 min. Subsequently, the reaction was terminated with 125 mM glycine for 5 min. After washing twice with PBS, it was centrifuged at 1000*g* to collect the cells. The fixed senescent MSCs were suspended

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in the cell lysis buffer [20 mM tris-HCl (pH 8.0), 85 mM KCl, and 0.5% NP-40]. After centrifugation at 1000*g*, the cell nuclei were precipitated and then resuspended in nuclear lysis buffer [50 mM tris-HCl (pH 8.0), 10 mM EDTA, and 1% SDS]. Chromatin fragments of 200 to 1000 bp were obtained by ultrasonic disruption and diluted with ChIP dilution buffer [16.7 mM tris-HCl, (pH 8.0), 1.2 mM EDTA, 0.01% SDS, 1% Triton X-100, and 167 mM NaCl]. We took 500 μg of chromatin, added antibodies (p300, SP1, and H3K27ac) and nonspecific IgG of the same species (table S4), incubated overnight at 4°C, and retained 50 μg of chromatin as the input control group. Subsequently, Protein A/G magnetic beads were added and incubated at 4°C for 2 hours to capture the antibody/DNA complex. After cleaning the magnetic beads with the washing buffer to remove nonspecific bindings, we added the decross-linking buffer containing protease K and incubate overnight. Last, DNA was extracted and purified by the phenol-chloroform method. The obtained purified DNA was used for the subsequent qPCR analysis. The percentage of input method is used for quantification and analysis. The specific primer sequences for ChIP-qPCR were listed in table S3.

Protein binding affinity prediction

We first applied MAPE-PPI with default parameters and pretrained model to predict which proteins are most likely to form complexes with p300. To further identify proteins that bind tightly to p300, we built structural models of p300 in complex with each candidate protein using AlphaFold3 (*79*).

To further analyze proteins that interact strongly with p300, we used Rosetta (v 3.14) to quantify protein-protein interaction strength. Specifically, we first applied FastRelax to gently relax the input complexes (-nstruct 1), constraining both backbone and side chains to their starting coordinates (-relax:constrain_relax_to_start_coords, -relax:coord_constrain_sidechains) with a coordinate-constraint SD of 0.3 Å (-relax:coord_cst_stdev 0.3), and scored structures with ref2015 (-score:weights ref2015). We then ran InterfaceAnalyzer on the A_B interface, again using ref2015, enabling separated-state repacking (-pack_separated) and packing statistics (-compute_packstat). The analysis produced interface_score.sc and the following metrics: dG_cross, dG_cross/dSASA × 100, dG_separated, dG_separated/dSASA × 100, per_residue_energy_int, ΔG (REU), dSASA_hydrophobic, dSASA_int, dSASA_polar, hbond_E_fraction, hbonds_int, nres_all, nres_int, and sc_value. Candidates were ranked by a composite score, and the top-scoring proteins were retained.

Single-cell and spatial transcriptomics analysis

Published scRNA-seq data from human and spatial transcriptomics from AD mouse brain were collected from the public dataset (*51*, *52*). scRNA-seq processing was carried out in Seurat (v 4.3.0.1), and cell types were annotated with CellMarker2. Differential expression was performed in Seurat (*80*). Interactive plots were created with ShinyCell (v 2.1.0). For spatial transcriptomics analysis, brain regions were labeled with the Allen Brain Atlas and Seurat, and DEGs analysis followed the single-cell workflow. Spatial visualizations were produced in ggplot2 (v 3.5.1). All single-cell and spatial analyses were run in R (v 4.4.3).

Statistical analysis

All experiments were performed with at least three biological replicates to ensure consistency. The data were expressed as means ± SEM. Two-tailed Student's *t* test was used to compare two groups. Comparisons of more than two groups were assessed using a one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test. Statistical significance was set at a *P* value of less than 0.05. The analysis was conducted using GraphPad 8.0 software.

Supplementary Materials

The PDF file includes:

Figs. S1 to S14
Tables S1 to S4
Legends for data S1 to S10

Other Supplementary Material for this manuscript includes the following:

Data S1 to S10

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